Distribution Functions and Simulations with Plots

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Continuous Distribution Functions

Normal

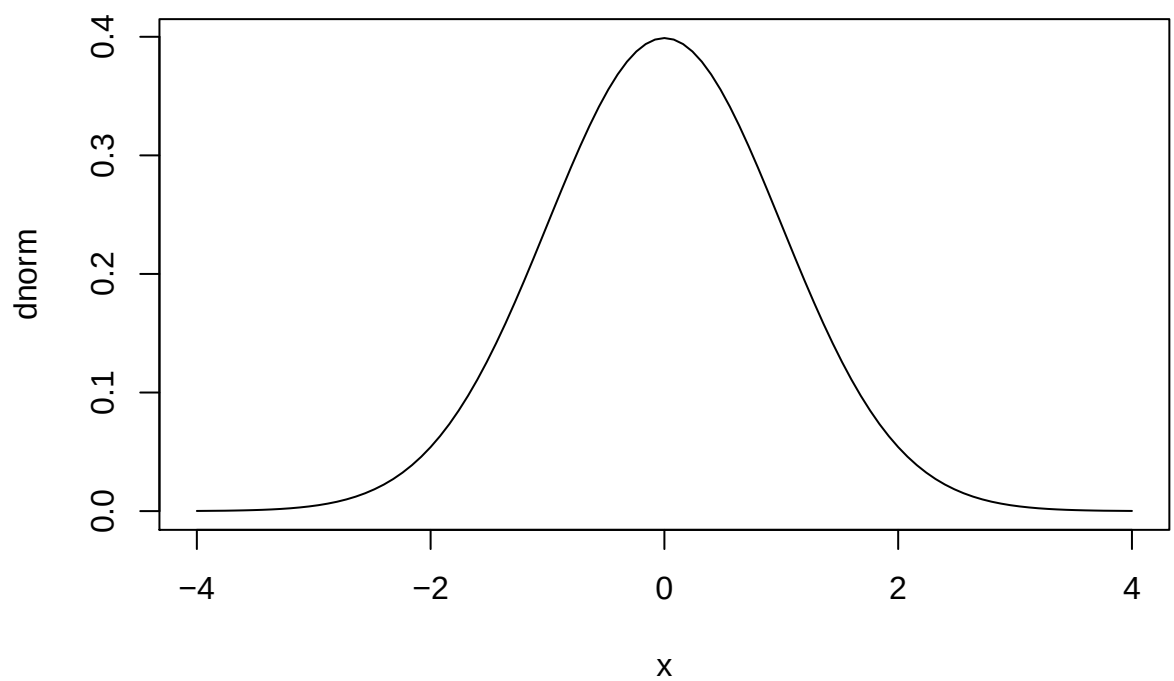
Generate 10 random numbers from a standard normal distribution

```
rnorm(10, 0, 1)
```

```
##  [1]  0.80866118 -0.47197456  1.91017018  0.82888715 -0.10313052
##  [6]  0.04261266 -0.90979265 -0.68095362  0.36332444  2.37066939
```

Make density plot for normal distribution (pdf)

```
plot(dnorm, from = -4, to = 4)
```




```
dnorm(0, 0, 1)
```

```
## [1] 0.3989423
```

Get values of normal cdf at point

```r
pnorm(0, 0, 1)
```

```
## [1] 0.5
```

Get x value for given probability

```r
qnorm(.5, 0, 1)
```

```
## [1] 0
```

Note: vectors can be used in first argument

```r
dnorm(c(-1,0,1),0, 1)
```

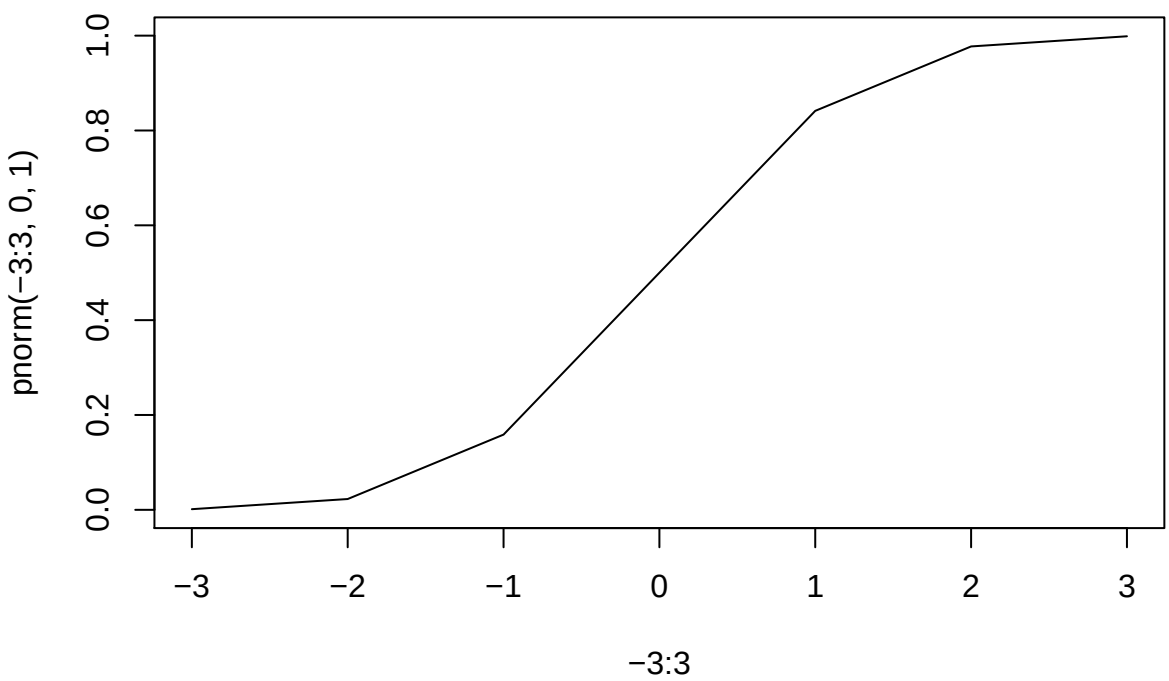
```
## [1] 0.2419707 0.3989423 0.2419707
```

```r
pnorm(c(-10,-2,0,2,10), 0, 1)
```

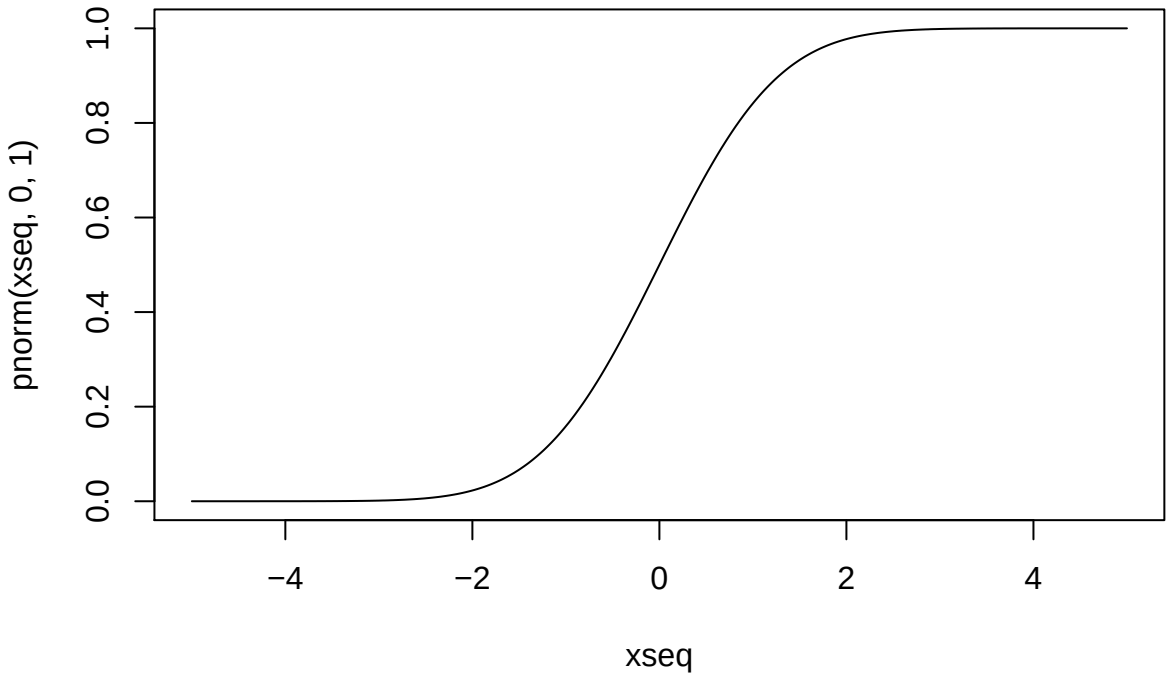
```
## [1] 7.619853e-24 2.275013e-02 5.000000e-01 9.772499e-01 1.000000e+00
```

Try to plot cdf

```r
plot(-3:3, pnorm(-3:3, 0, 1), type = 'l')
```



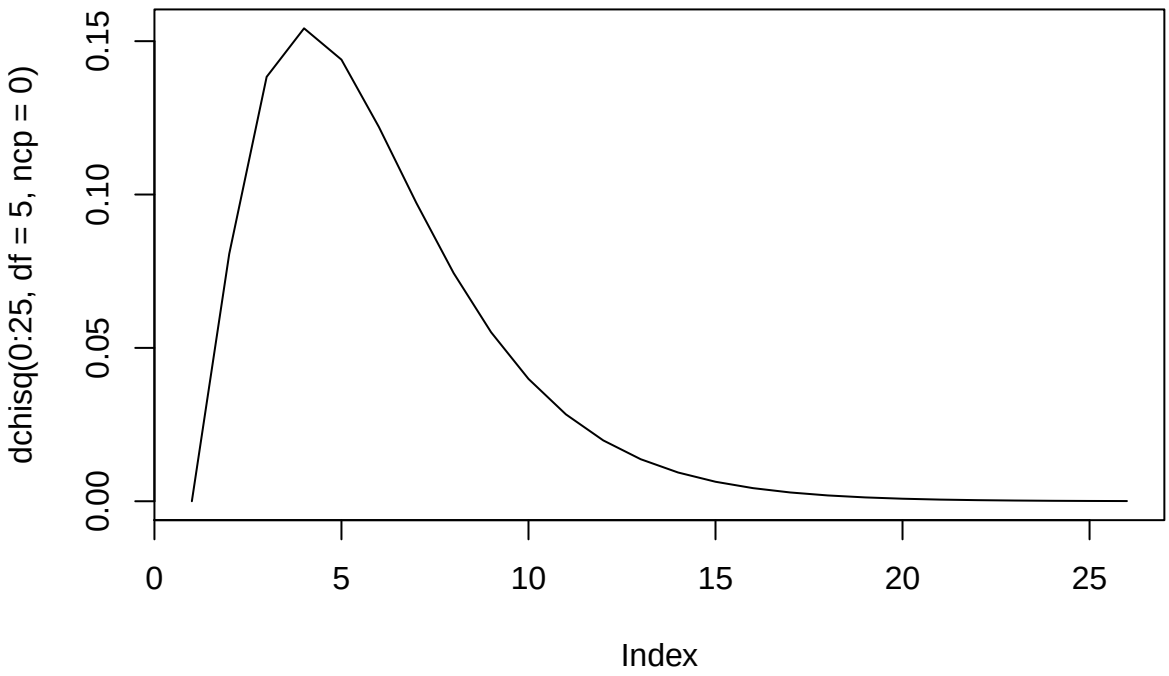

```r
xseq <- seq(-5, 5, .01)
plot(xseq, pnorm(xseq, 0, 1), type = 'l')
```



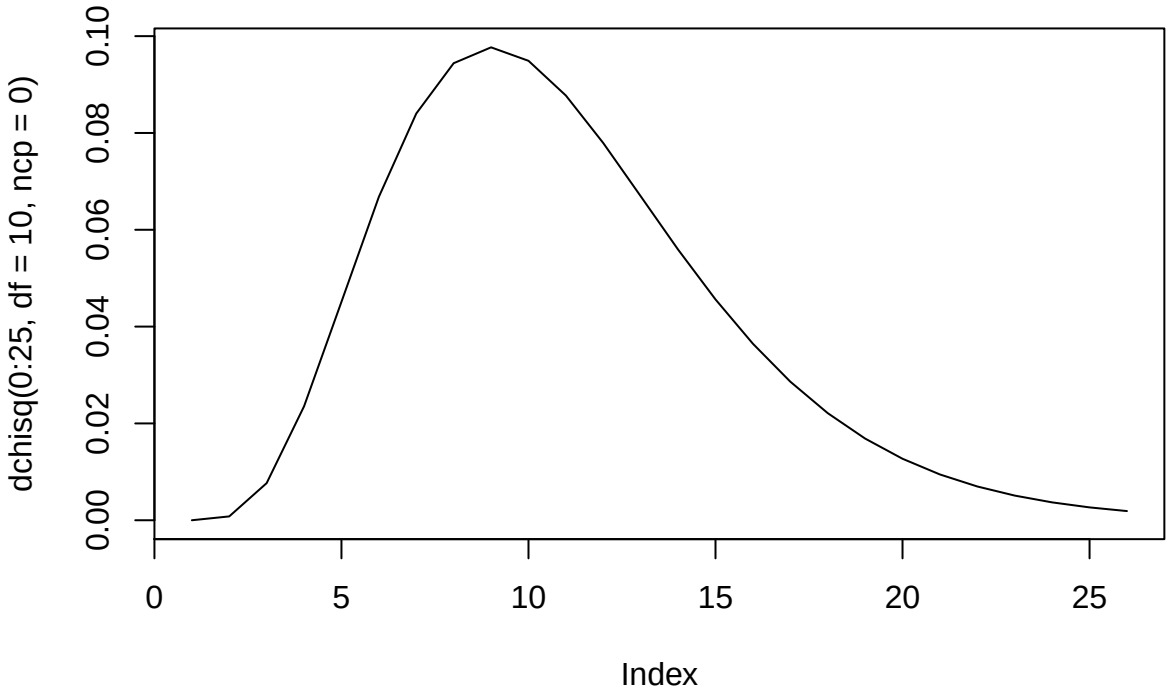
Chi-Squared

dchisq equivalent to dnorm, etc

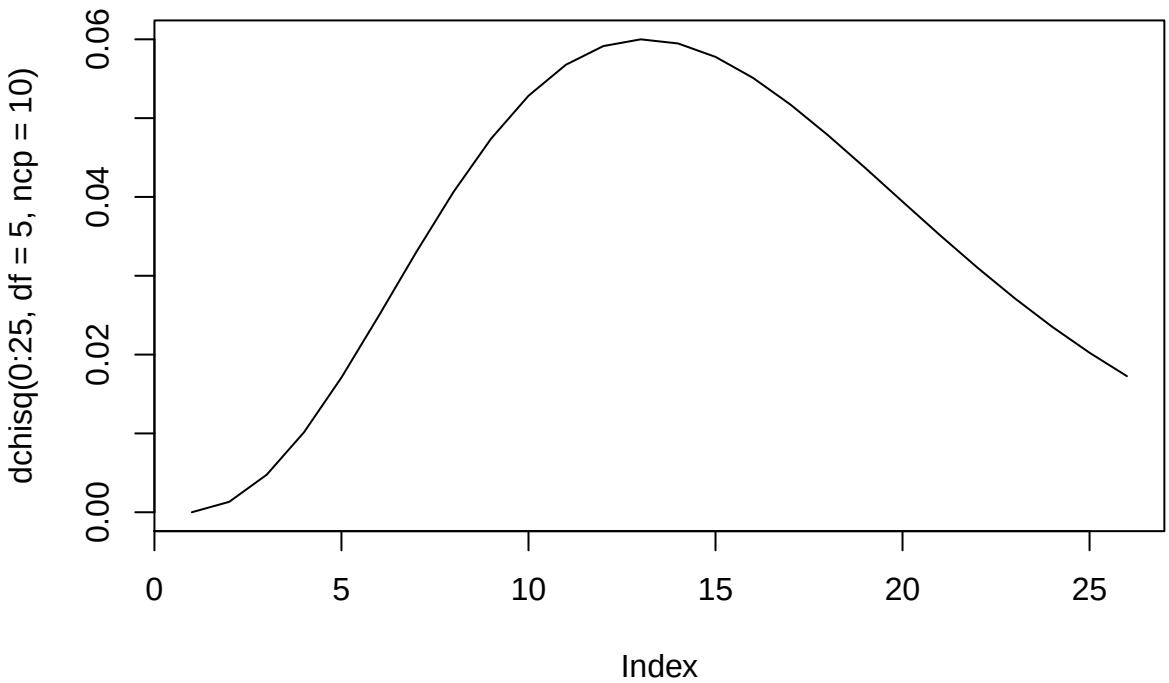
```
plot(dchisq(0:25, df = 5, ncp = 0), type = 'l') # 5 degrees of freedom
```




```r
plot(dchisq(0:25, df = 10, ncp = 0), type = 'l') # 5 degrees of freedom
```



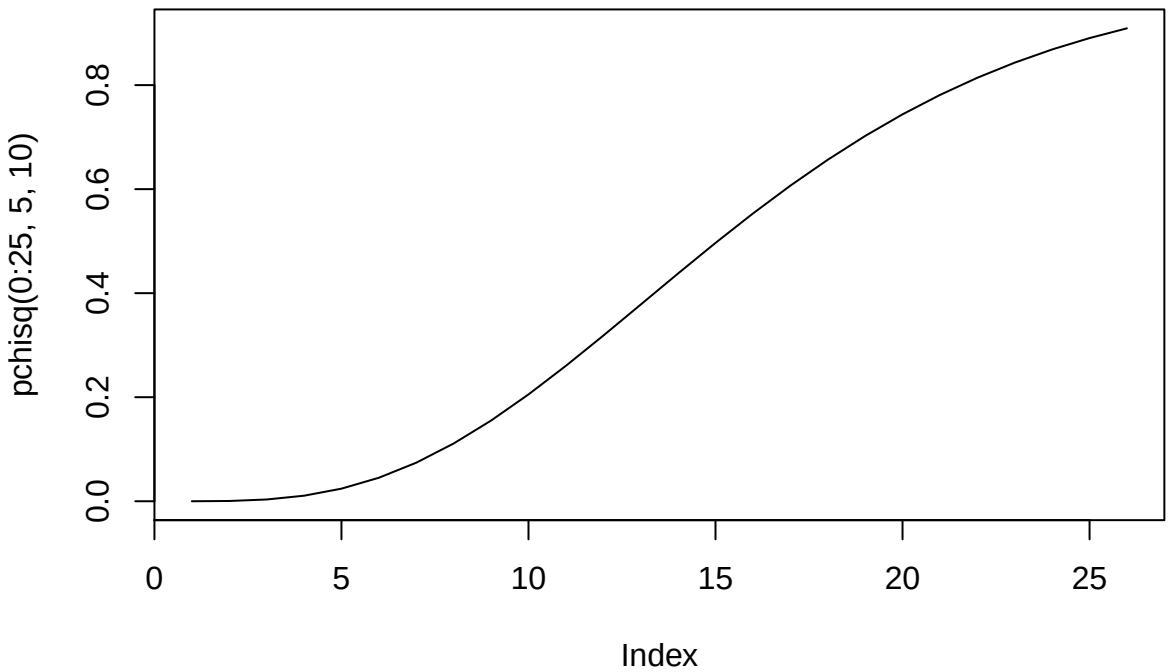

```r
plot(dchisq(0:25, df = 5, ncp = 10), type = 'l') # 5 degrees of freedom
```




```r
dchisq(10, 5, 10)
```

```
## [1] 0.05677048
```

```r
plot(pchisq(0:25 ,5 ,10), type = 'l')
```




```r
pchisq(10, 5, 10)
```

```
## [1] 0.2603024
```

```r
qchisq(p =.5, 5, 10)
```

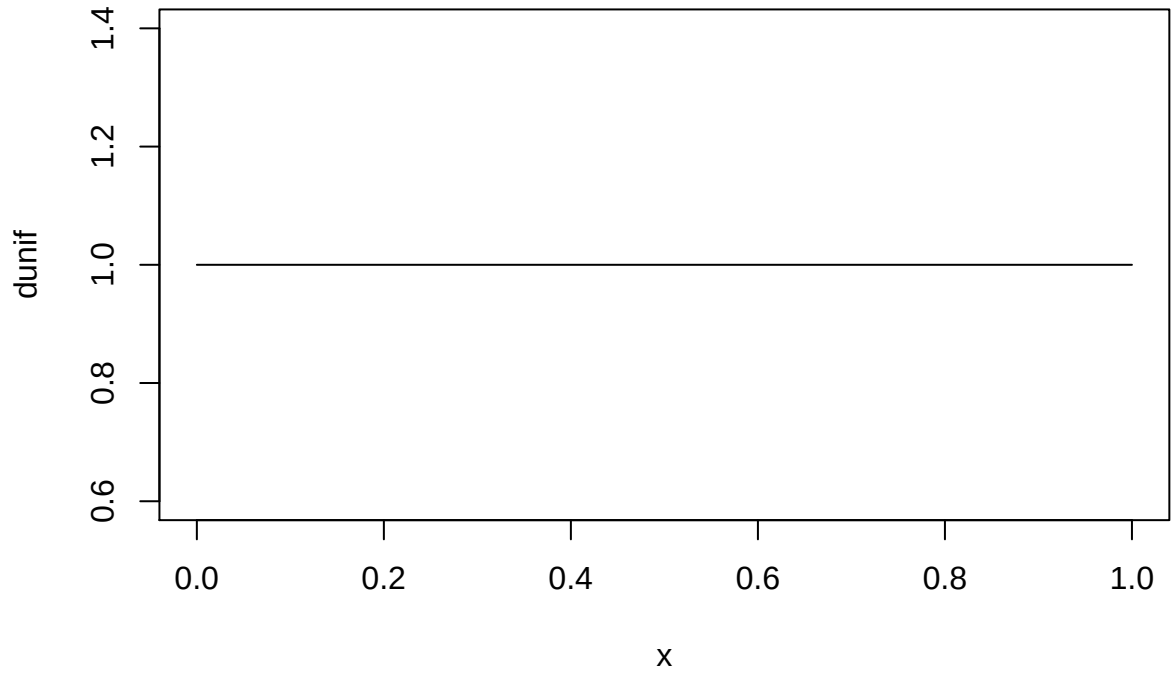
```
## [1] 14.05841
```

```r
rchisq(n = 10, 5, 10)
```

```
##  [1] 19.180248 14.310605  8.914507 21.174312 19.958243 11.632980 24.726473
##  [8] 22.472686 18.899318 22.685897
```

Uniform

```r
plot(dunif)
```




```r
dunif(.5, min = 0, max = 1)
```

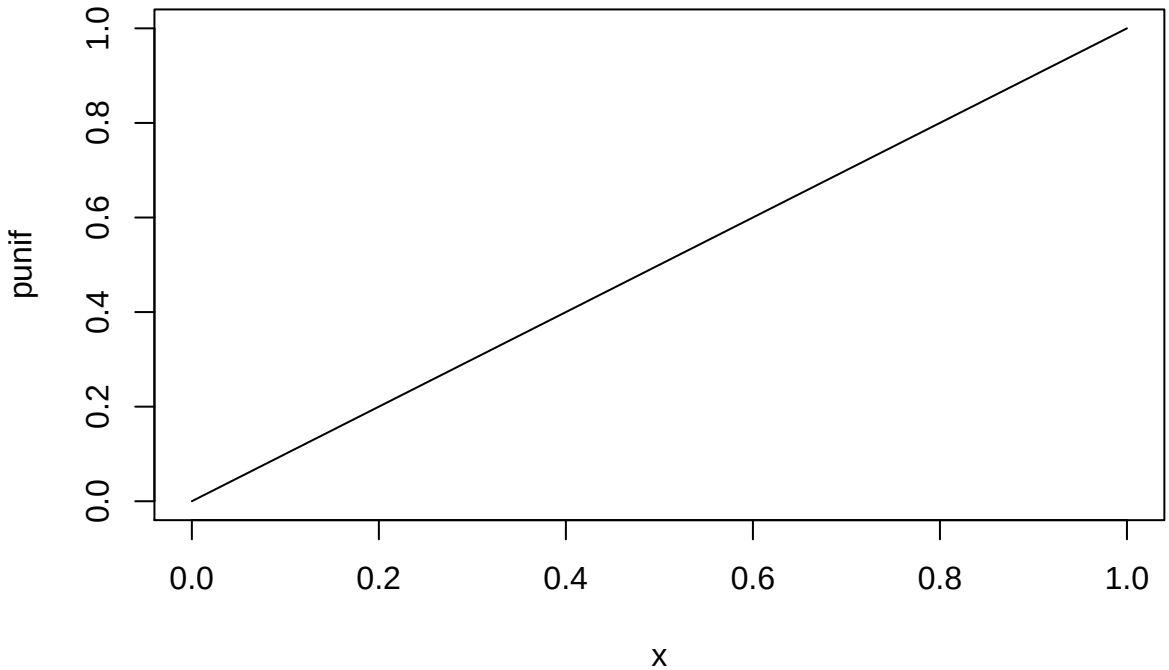
```
## [1] 1
```

Checking for value outside of min and max range will return 0

```r
dunif(1.5, 0, 1)
```

```
## [1] 0
```

```r
plot(punif)
```




```r
punif(.5, 0, 1)
```

```
## [1] 0.5
```

```r
punif(.8, 0, 1)
```

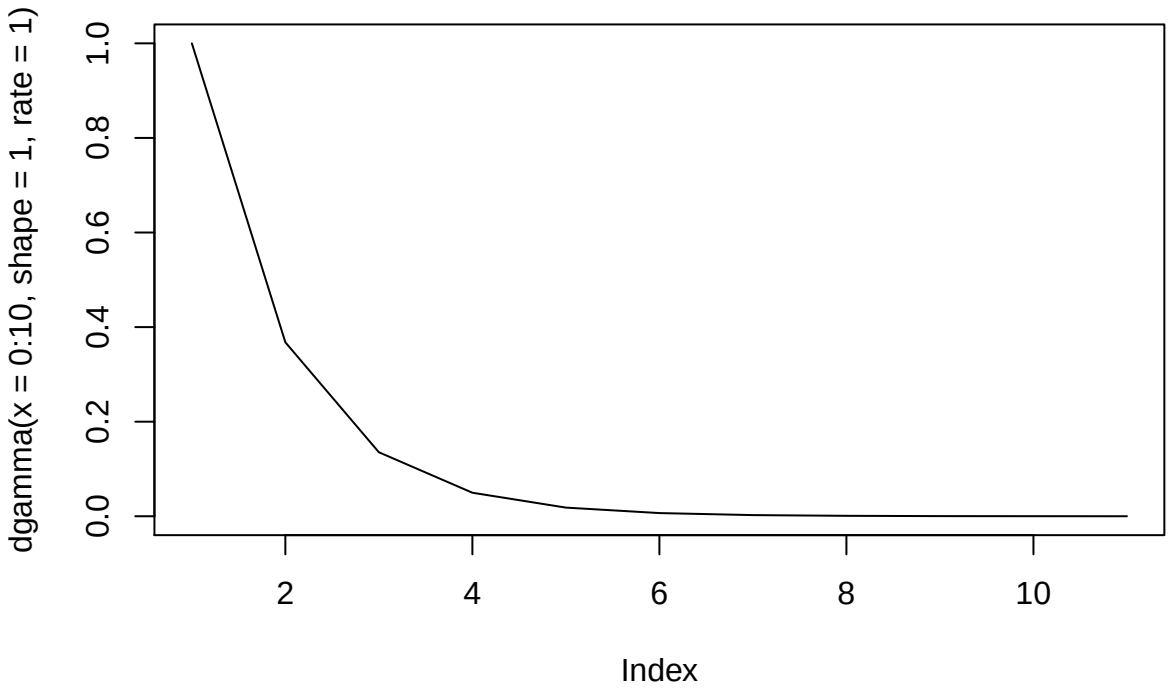
```
## [1] 0.8
```

```r
runif(10, 0, 1)
```

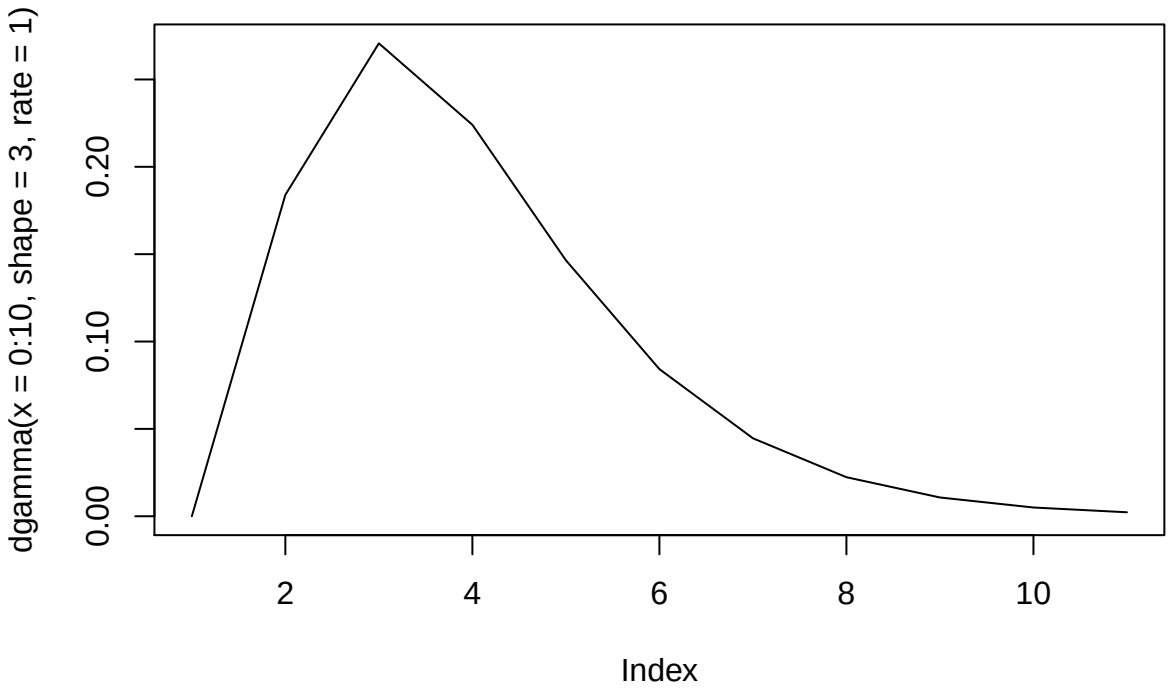
```
##  [1] 0.06608931 0.67357512 0.40310591 0.48819537 0.03538311 0.09099916
##  [7] 0.04723176 0.29622046 0.70356940 0.09638753
```

Gamma

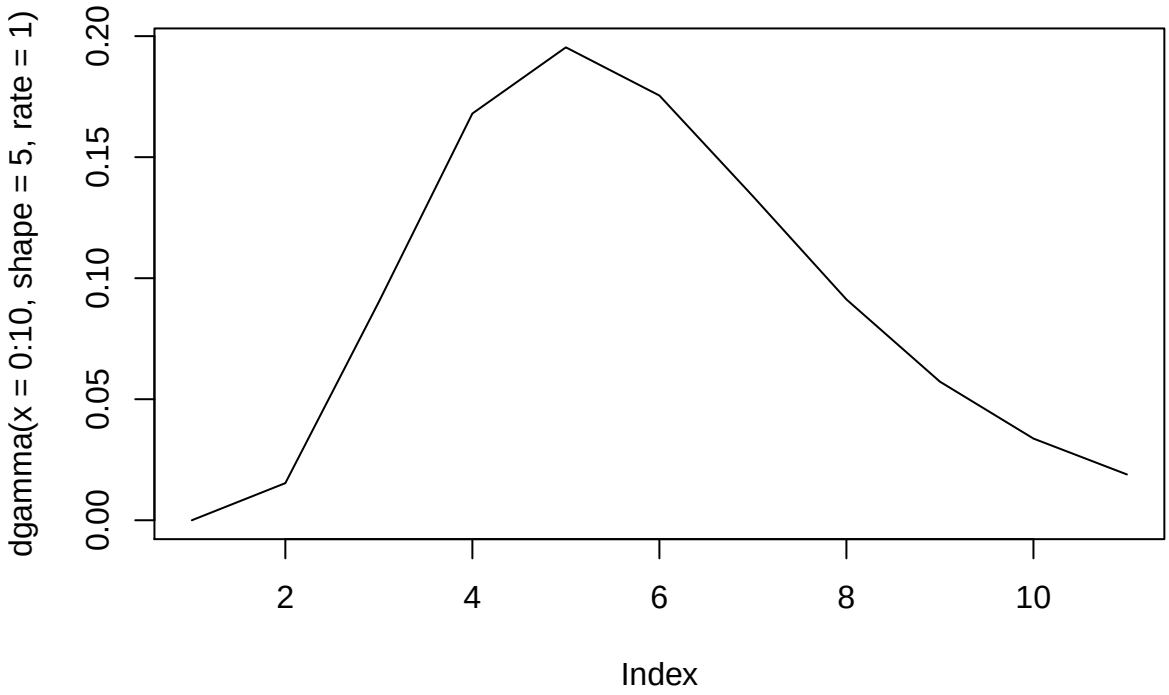
```r
plot(dgamma(x = 0:10, shape = 1, rate = 1 ),type = 'l')
```



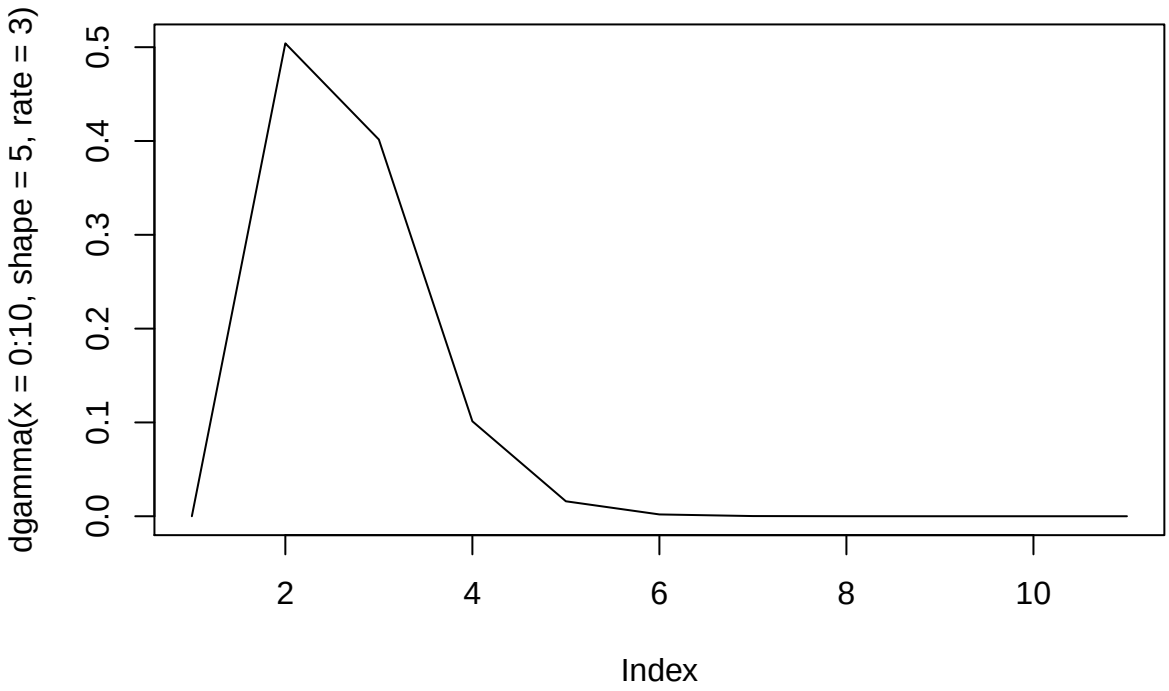

```
plot(dgamma(x = 0:10, shape = 3, rate = 1 ),type = 'l')
```




```
plot(dgamma(x = 0:10, shape = 5, rate = 1 ),type = 'l')
```



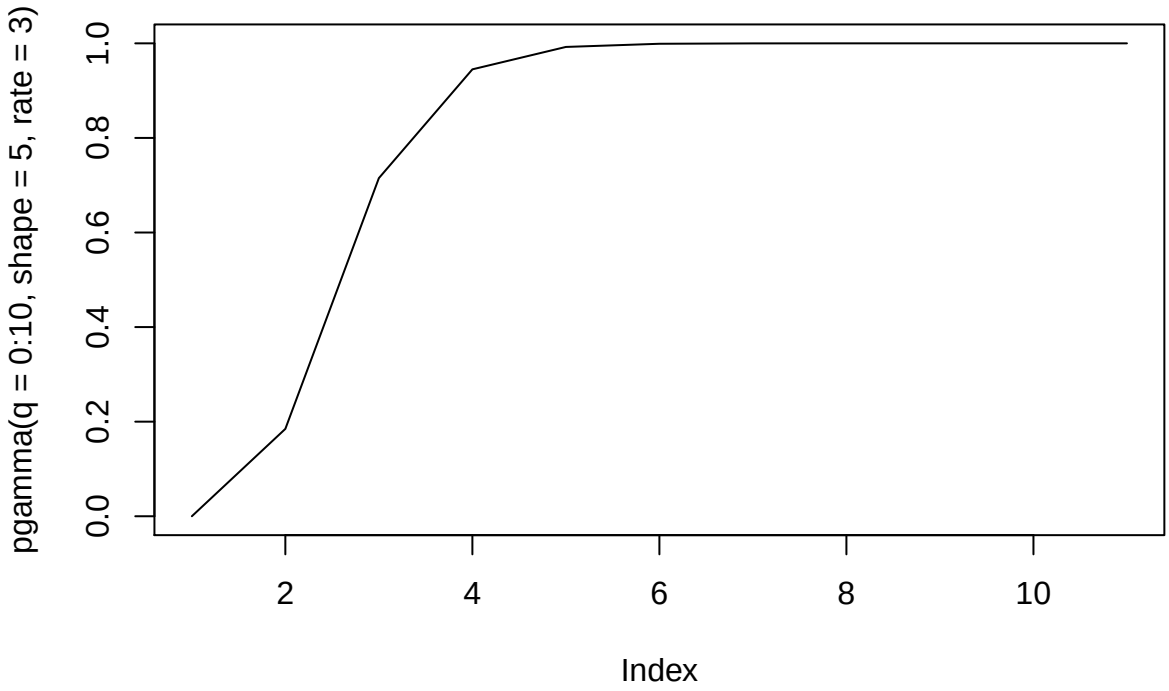

```
plot(dgamma(x = 0:10, shape = 5, rate = 3 ),type = 'l')
```




```r
dgamma(5, shape = 5, rate = 3)
```

```
## [1] 0.001935788
```

```r
plot(pgamma(q = 0:10, shape = 5, rate = 3), type = 'l')
```



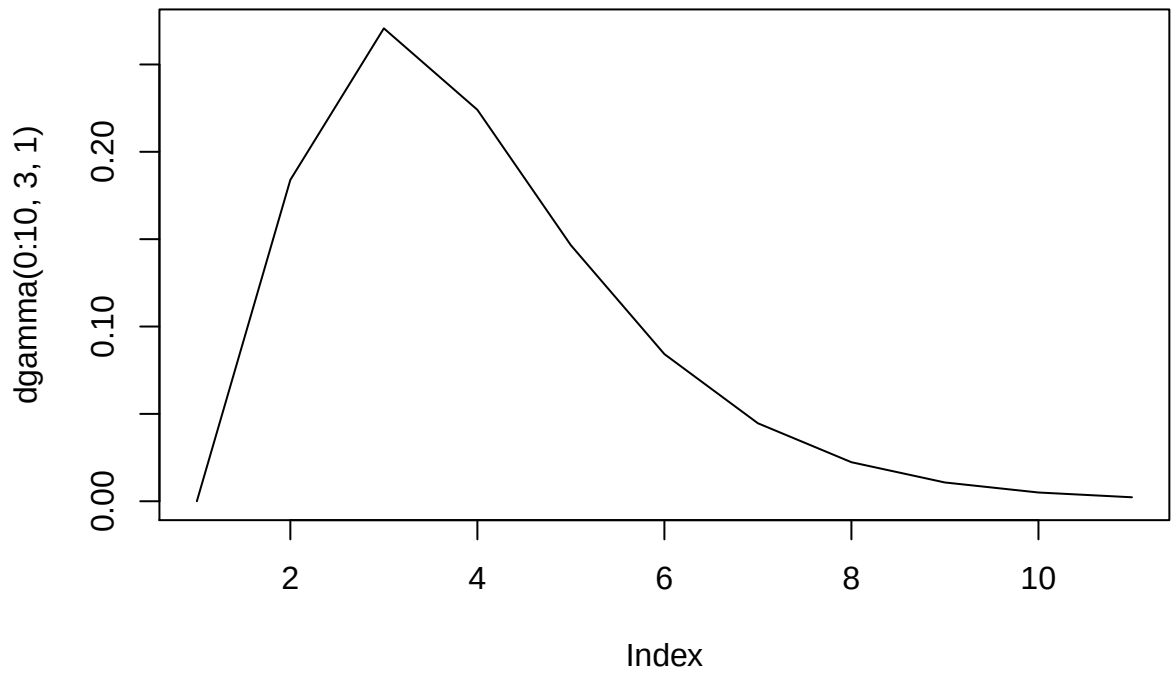

```r
pgamma(2, 5, 3)
```

```
## [1] 0.7149435
```

```r
qgamma(.5, 5, 3)
```

```
## [1] 1.55697
```

```r
plot(dgamma(0:10, 3, 1), type = 'l')
```

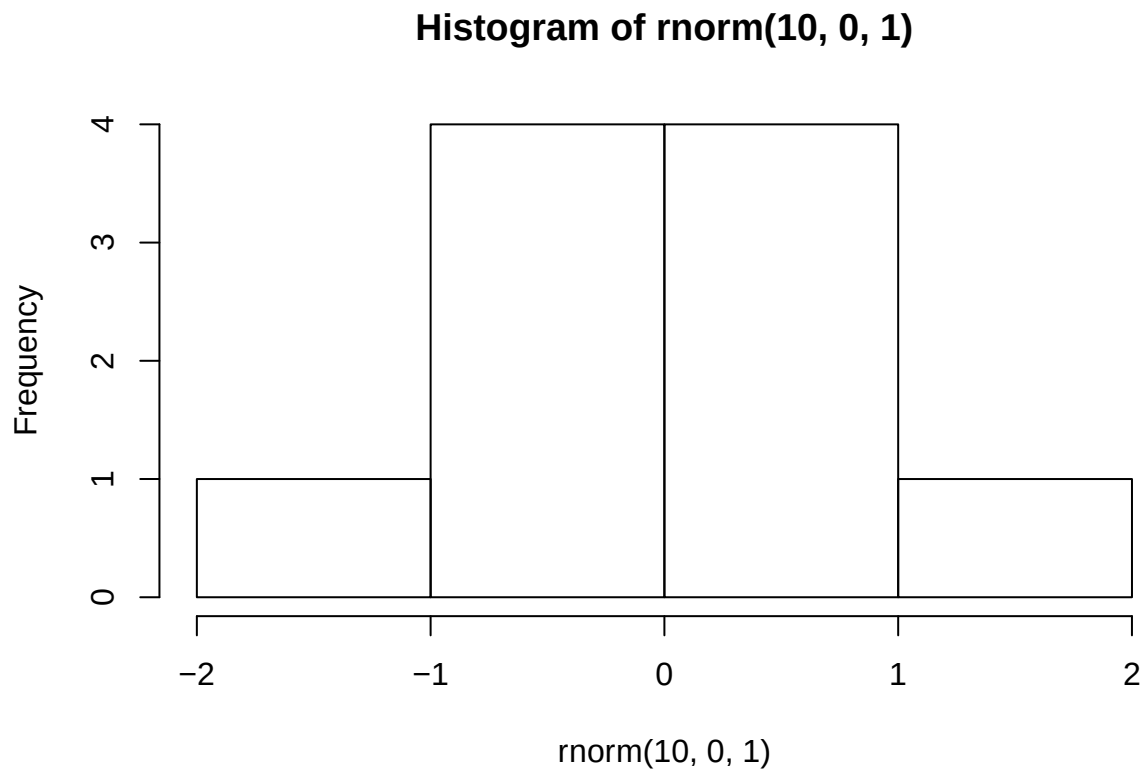



```r
rgamma(10, 3, 1)
```

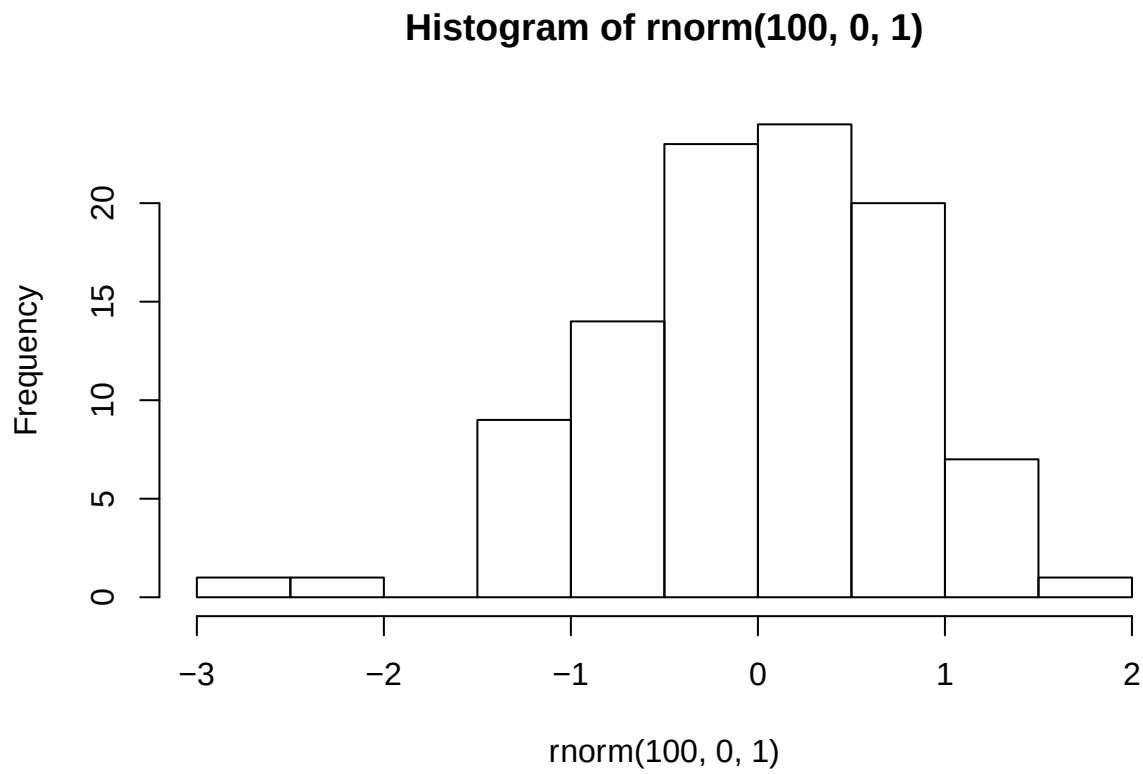
```
##  [1] 1.719964 4.146774 3.865515 3.152447 3.036109 6.112558 1.504726
##  [8] 1.460405 3.080376 2.826879
```

Bonus plots

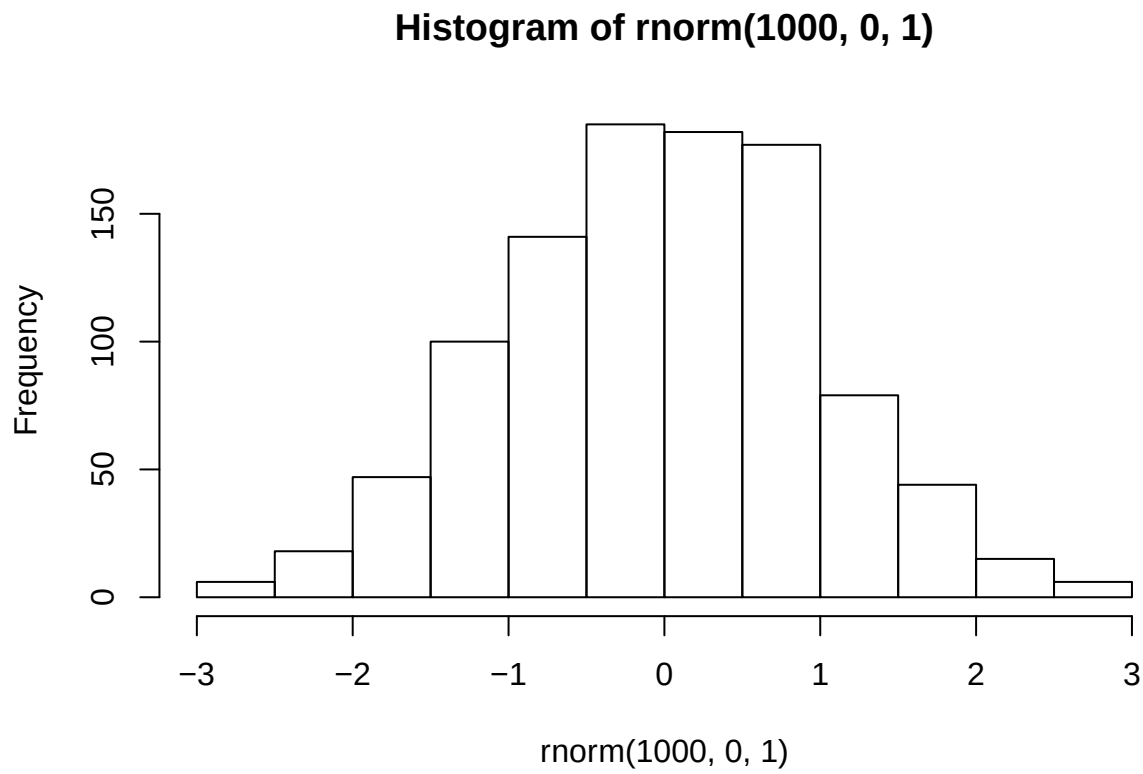
```r
hist(rnorm(10, 0, 1))
```



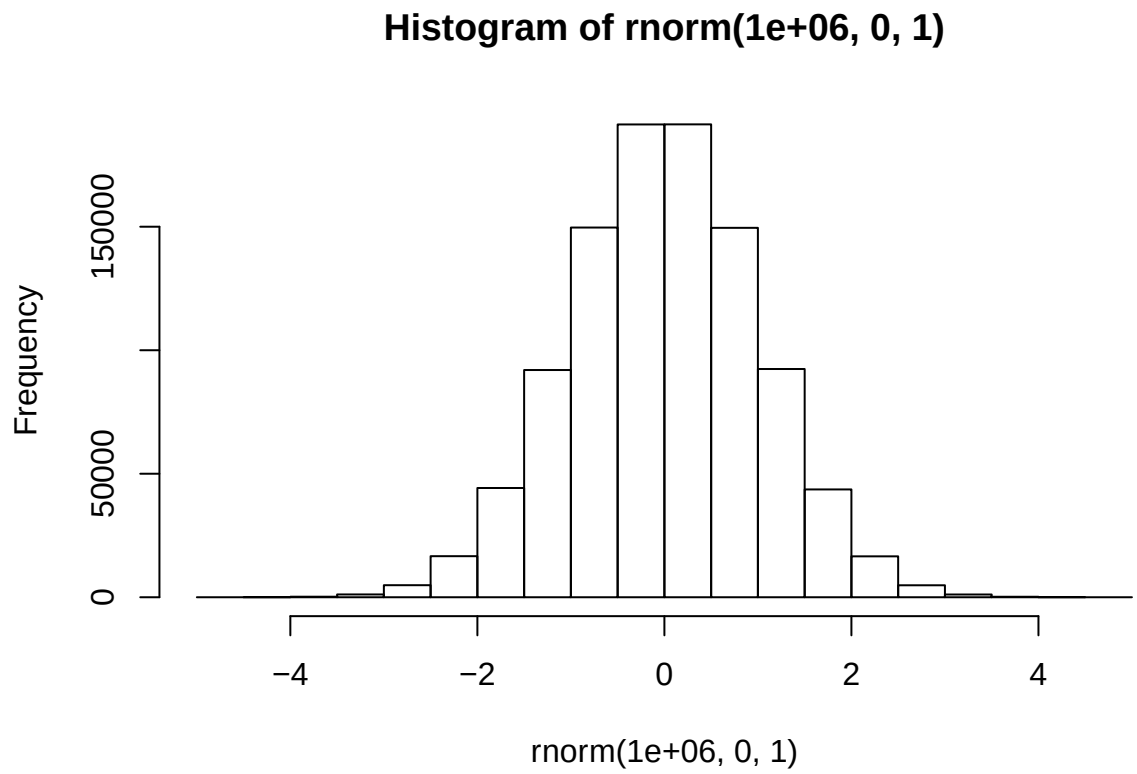

```r
hist(rnorm(100, 0, 1))
```




```
hist(rnorm(1000, 0, 1))
```



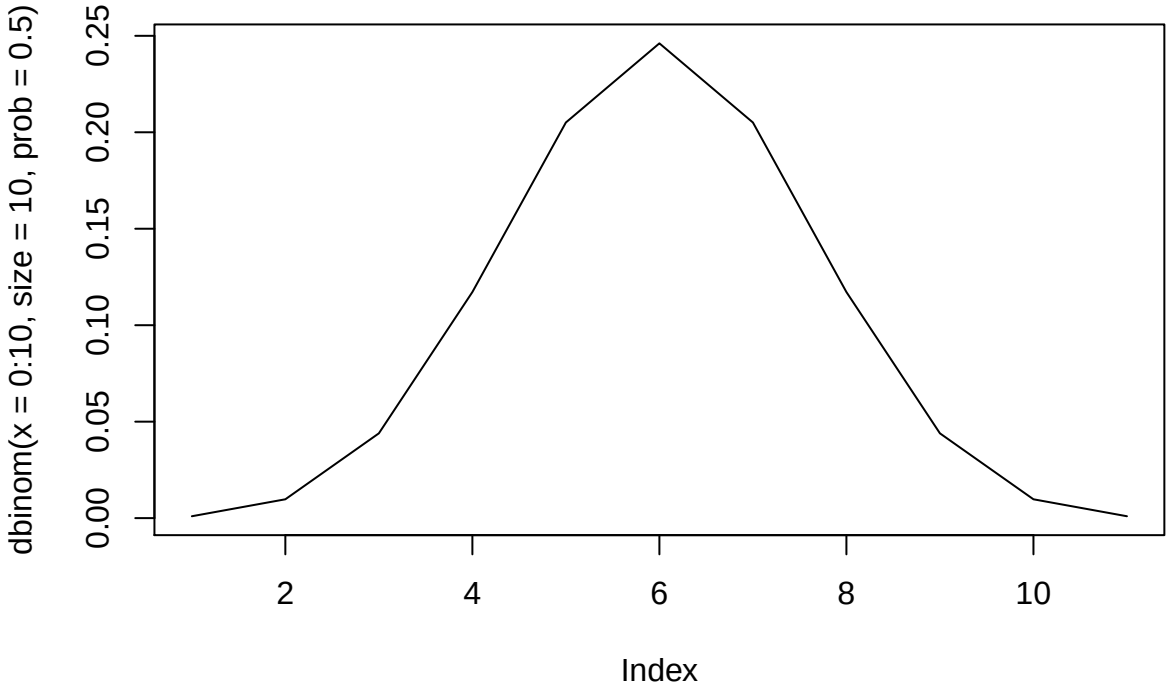

```r
hist(rnorm(1000000, 0, 1))
```



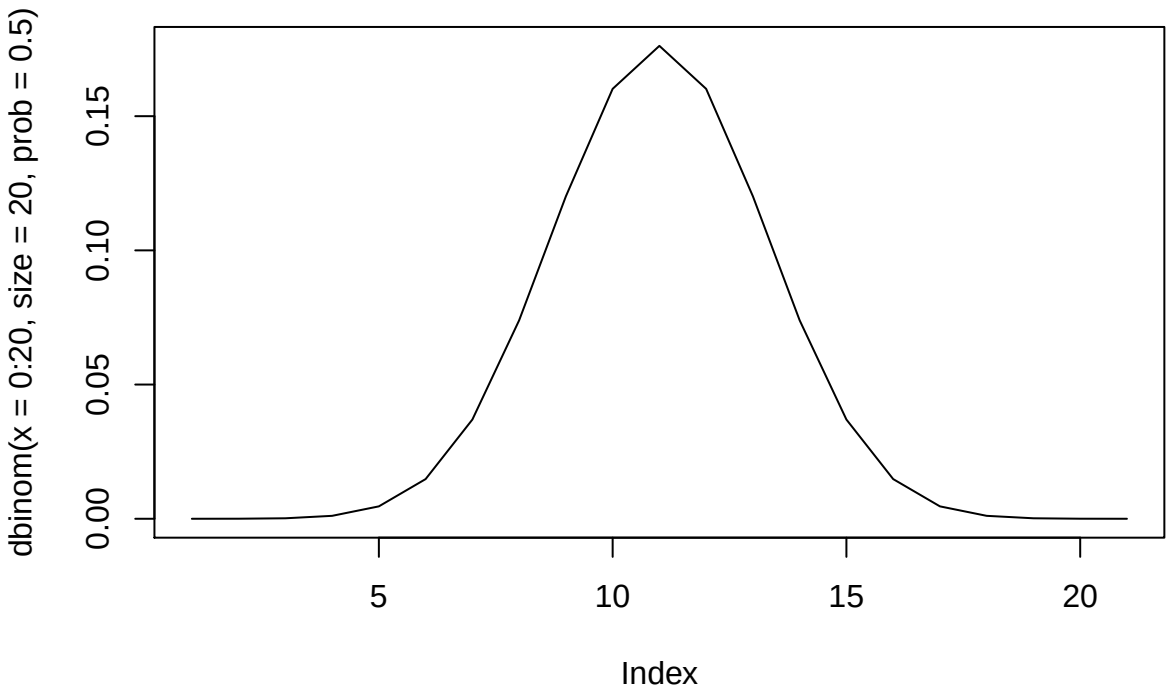
Discrete Distribution Functions

Binomial

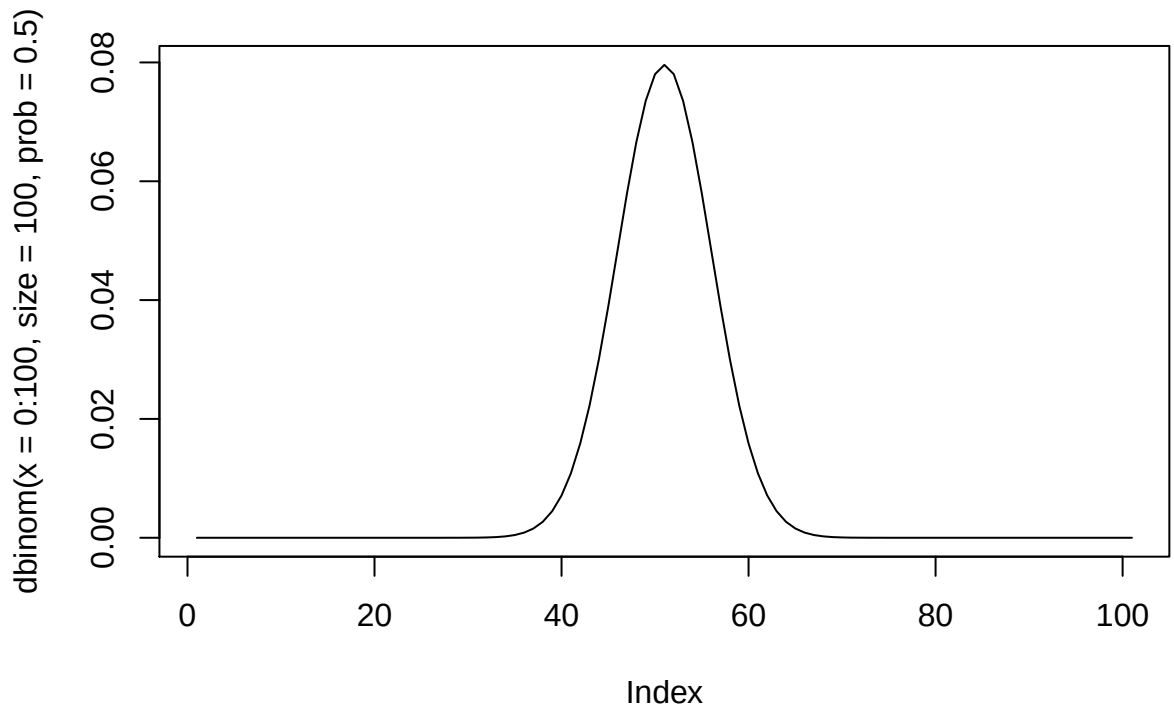
```
plot(dbinom(x = 0:10, size = 10, prob = .5), type = 'l')
```


dbinom(x = 0:10, size = 10, prob = 0.5)
0.00
0.05
0.10
0.15
0.20
0.25
2
4
6
8
10
Index

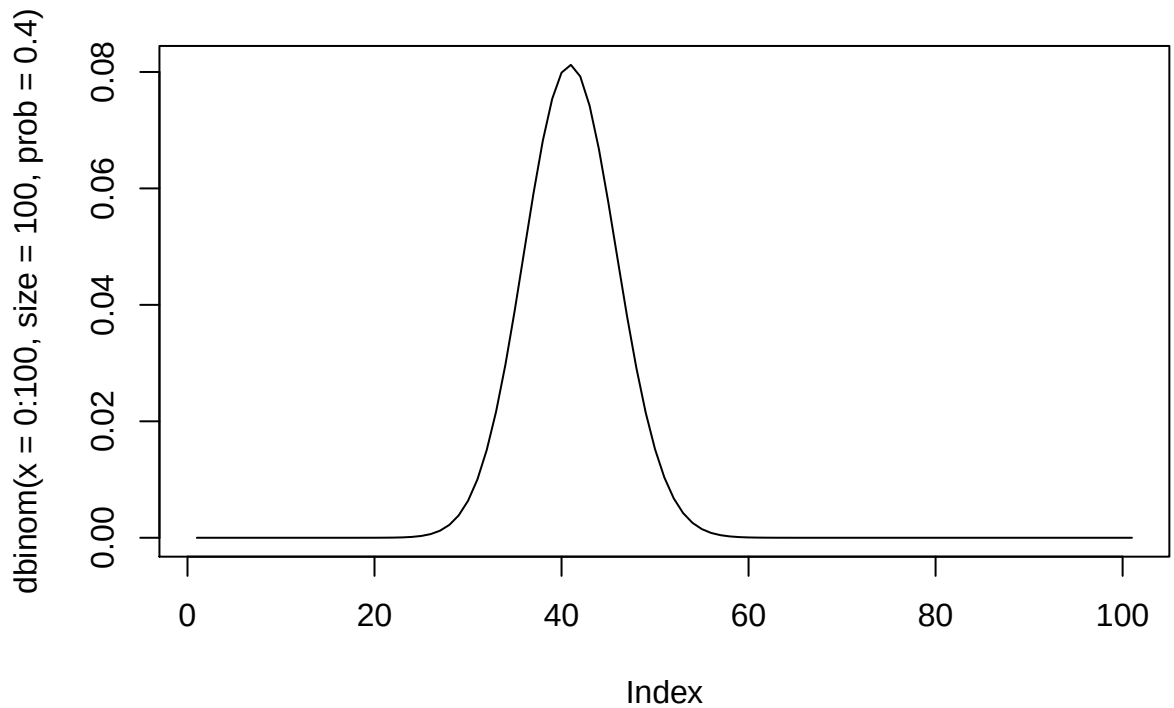
```r
plot(dbinom(x = 0:20, size = 20, prob = .5), type = 'l')
```



```r
plot(dbinom(x = 0:100, size = 100, prob = .5), type = 'l')
```



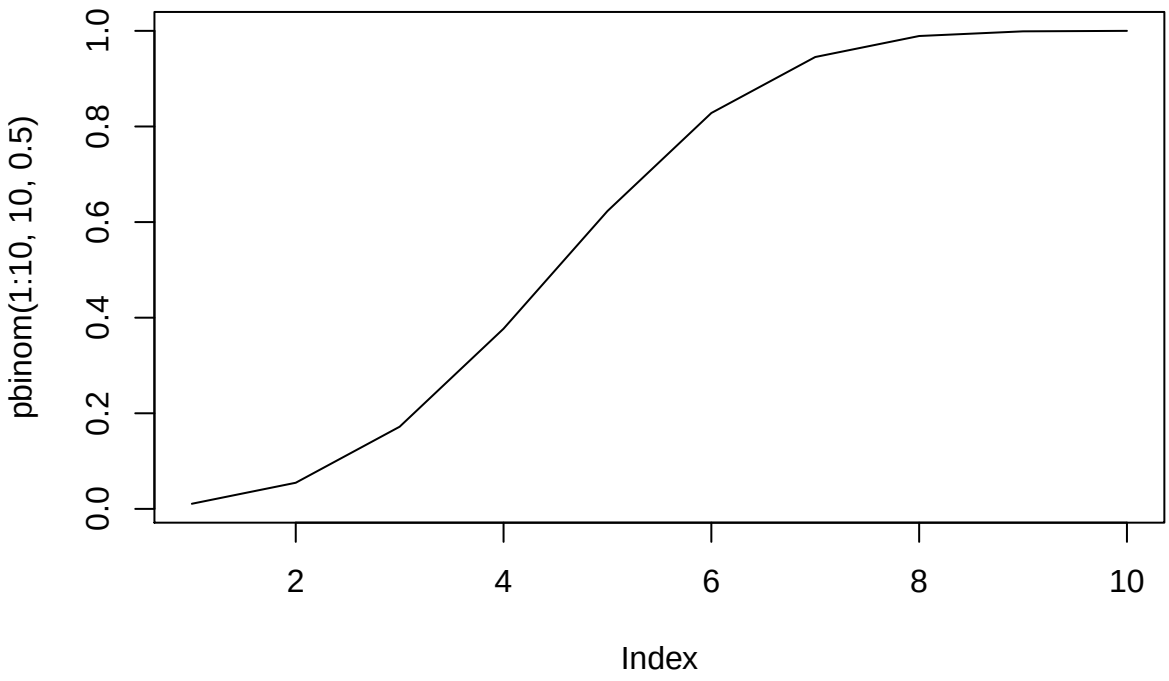
```r
plot(dbinom(x = 0:100, size = 100, prob = .4), type = 'l')
```



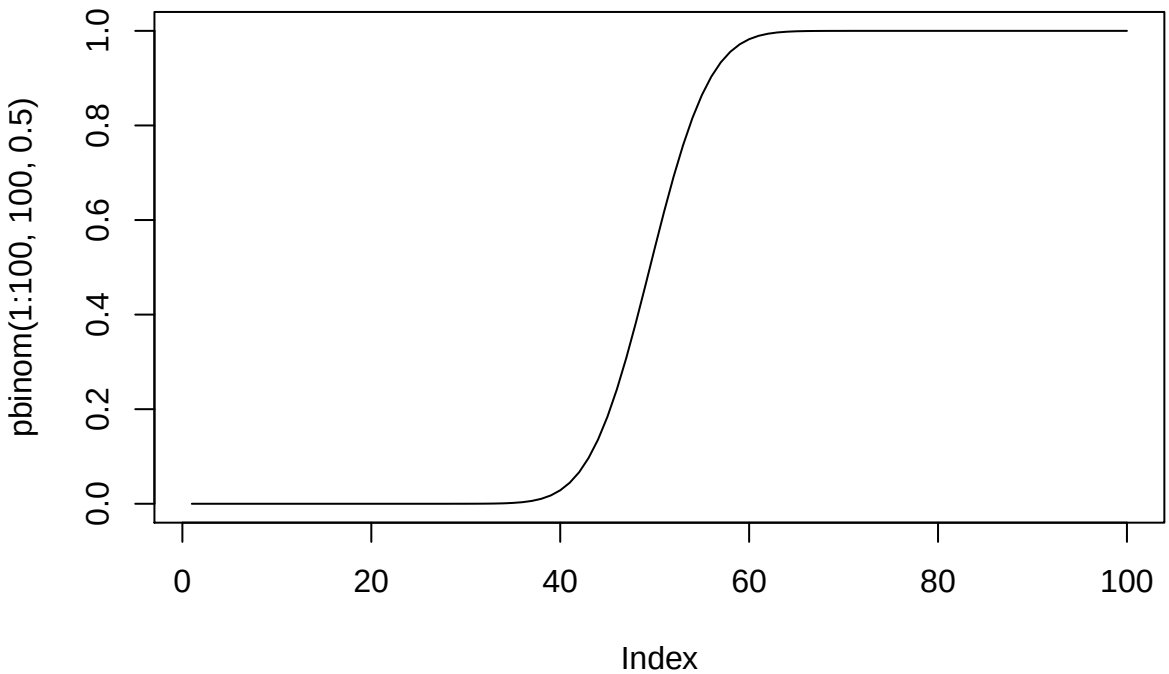

```r
dbinom(0:10, size = 10, prob = .5)
```

```
##  [1] 0.0009765625 0.0097656250 0.0439453125 0.1171875000 0.2050781250
##  [6] 0.2460937500 0.2050781250 0.1171875000 0.0439453125 0.0097656250
## [11] 0.0009765625
```

```r
plot(pbinom(1:10, 10, .5), type = 'l')
```



```r
plot(pbinom(1:100, 100, .5), type = 'l')
```




```r
qbinom(.5, 100, .5)
```

```
## [1] 50
```

```r
rbinom(10, 100, .5)
```

```
##  [1] 57 44 52 44 47 39 54 50 51 50
```

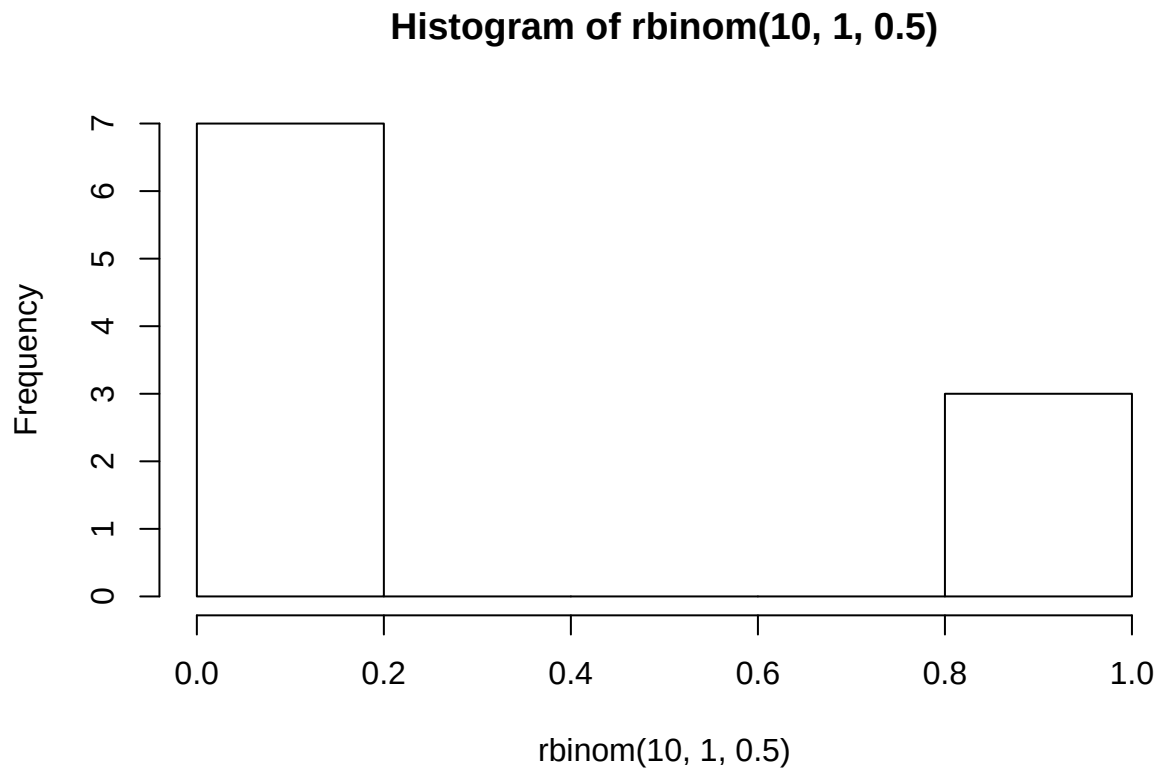
Simulate coin flip

```r
rbinom(100, 1, .5)
```

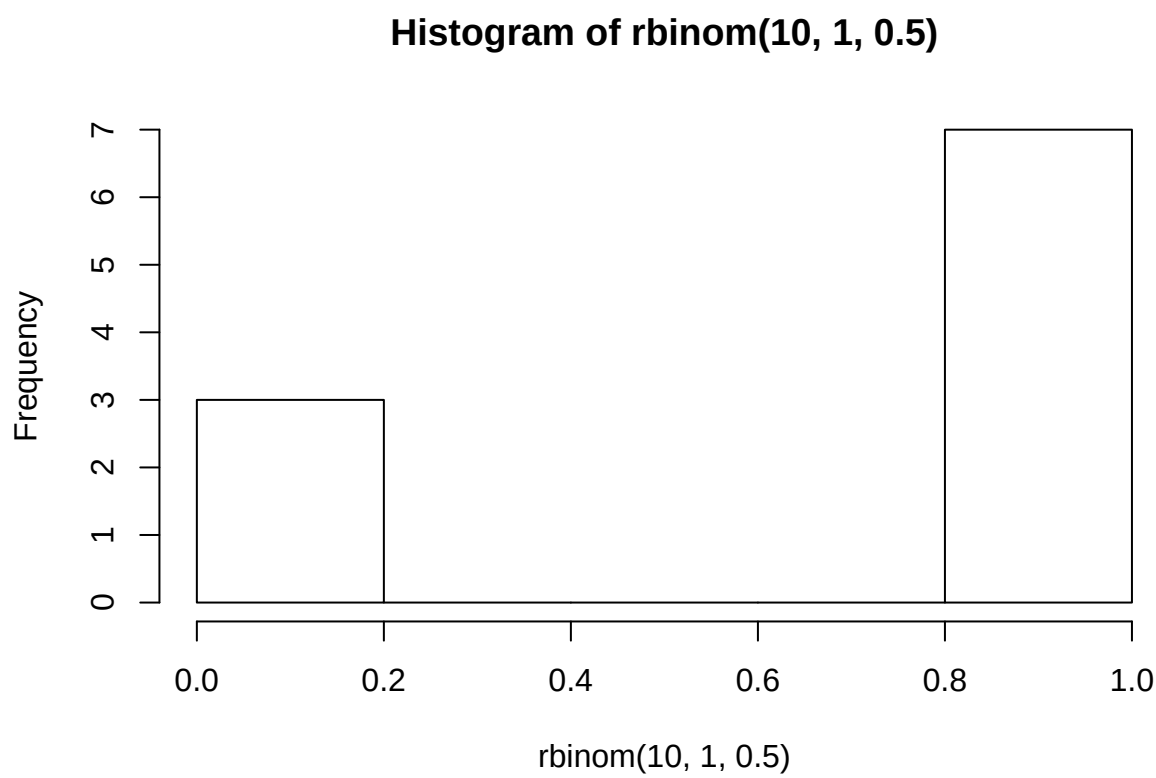
```
##   [1] 0 0 1 1 0 0 1 0 0 0 0 0 1 1 0 0 0 0 0 0 1 0 0 1 1 1 1 0 1 0 0 1 0 1 1
##  [36] 1 0 1 1 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 1 0 0 1 1 1 0 0 1 0 1 0 1 1 1 1
##  [71] 0 1 1 1 1 1 0 0 1 0 0 0 1 0 0 1 0 1 1 1 0 0 0 1 1 1 1 1 0 1
```

Re-simulate a few times to see the variation at a small n

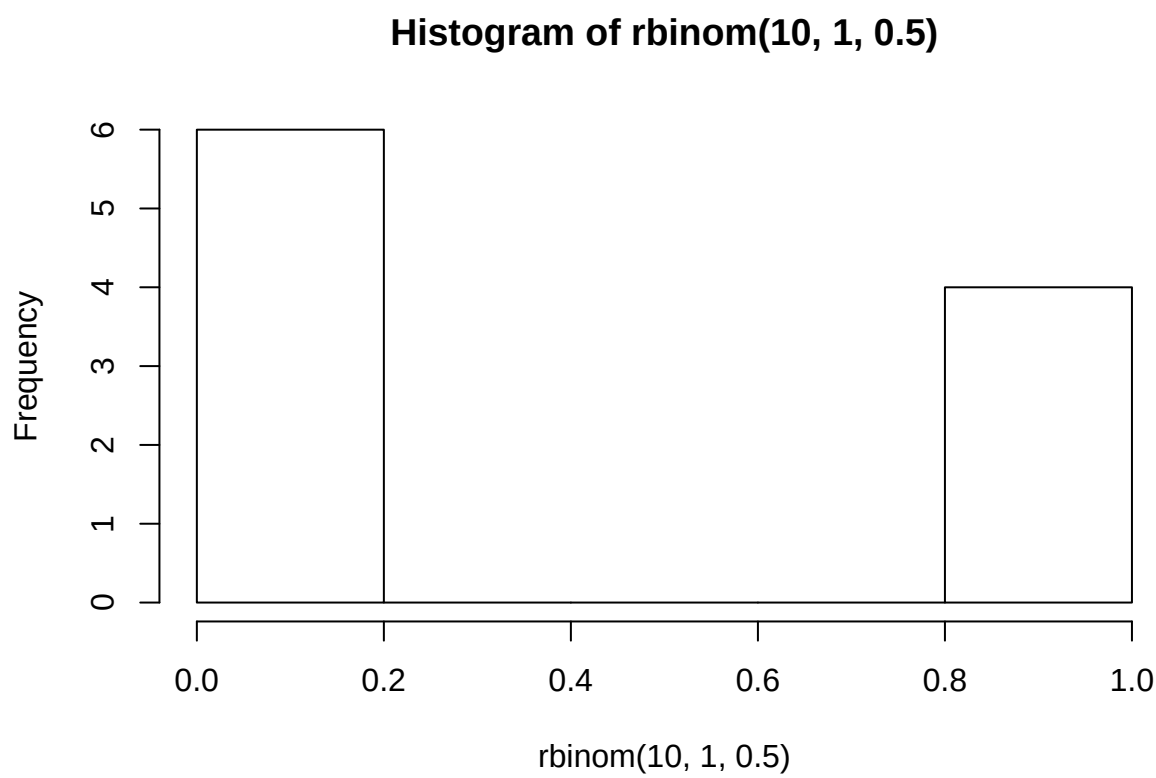
```
hist(rbinom(10, 1, .5))
```



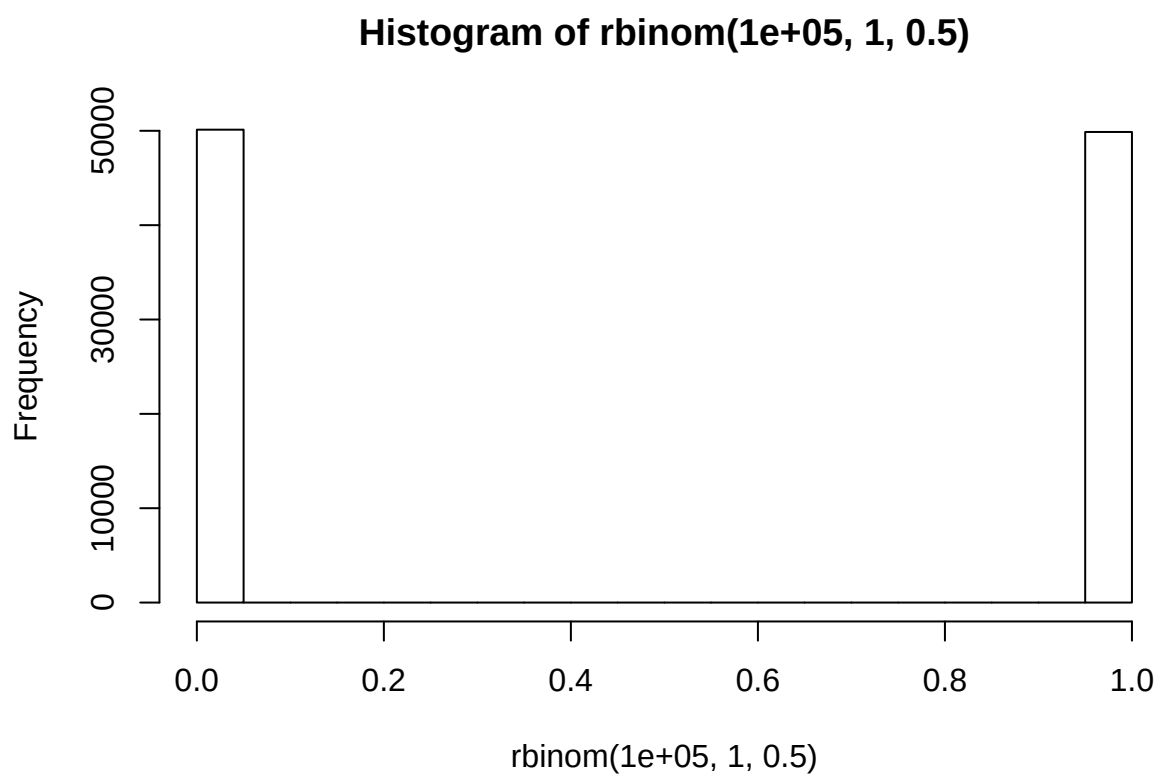
```r
hist(rbinom(10, 1, .5))
```



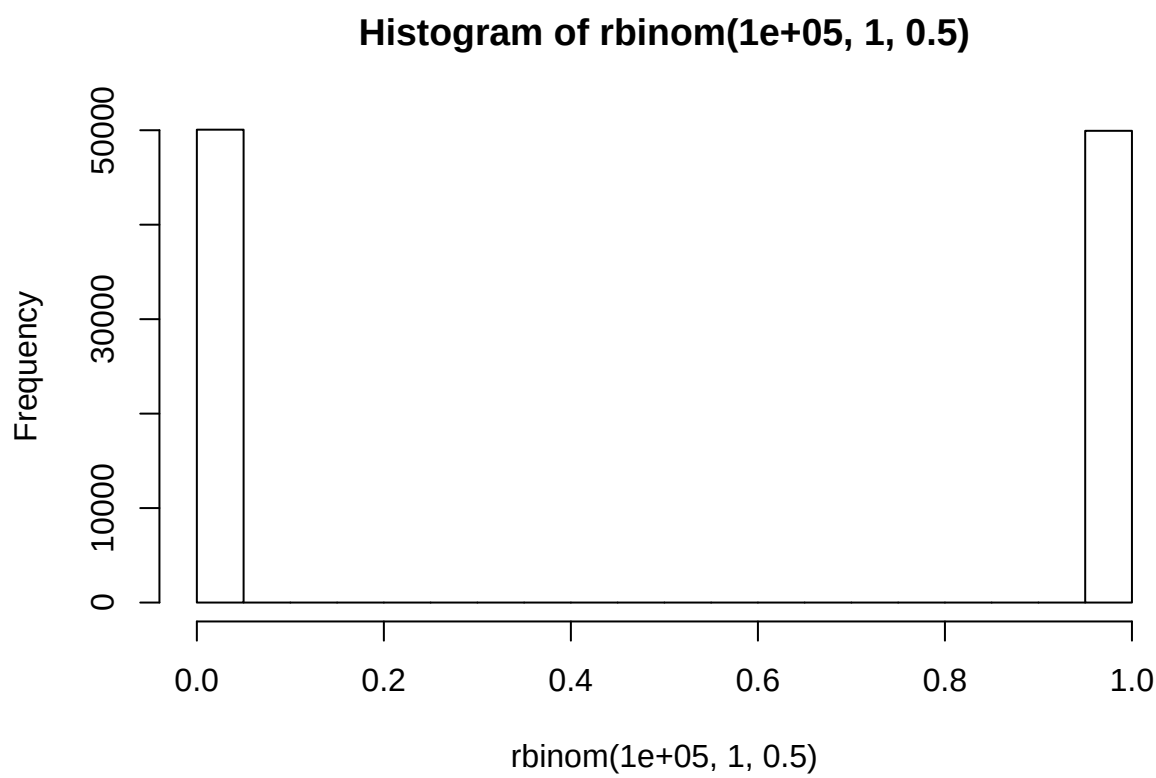
```r
hist(rbinom(10, 1, .5))
```



```r
hist(rbinom(100000, 1, .5))
```

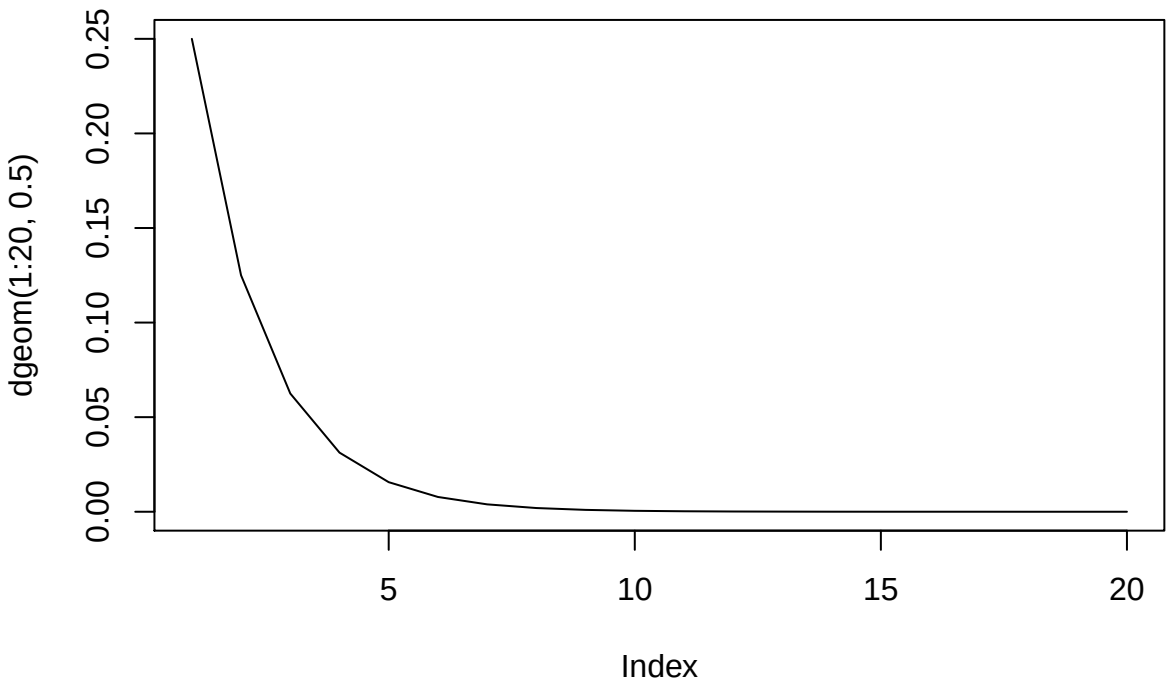


```r
hist(rbinom(100000, 1, .5))
```

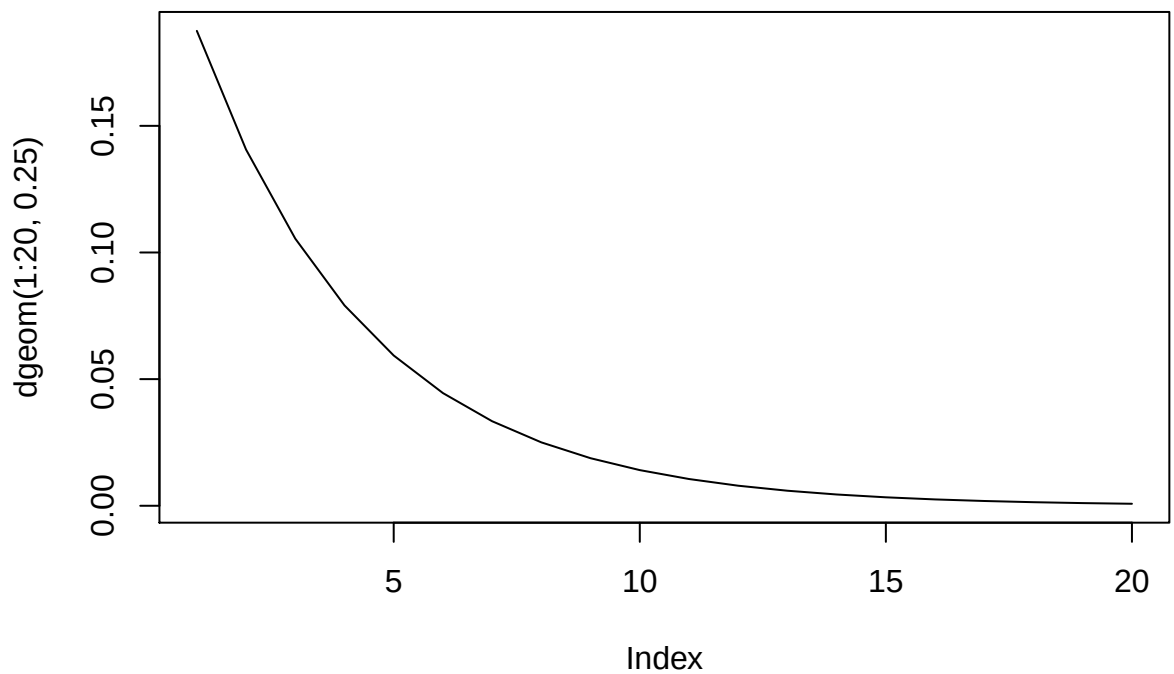


Geometric

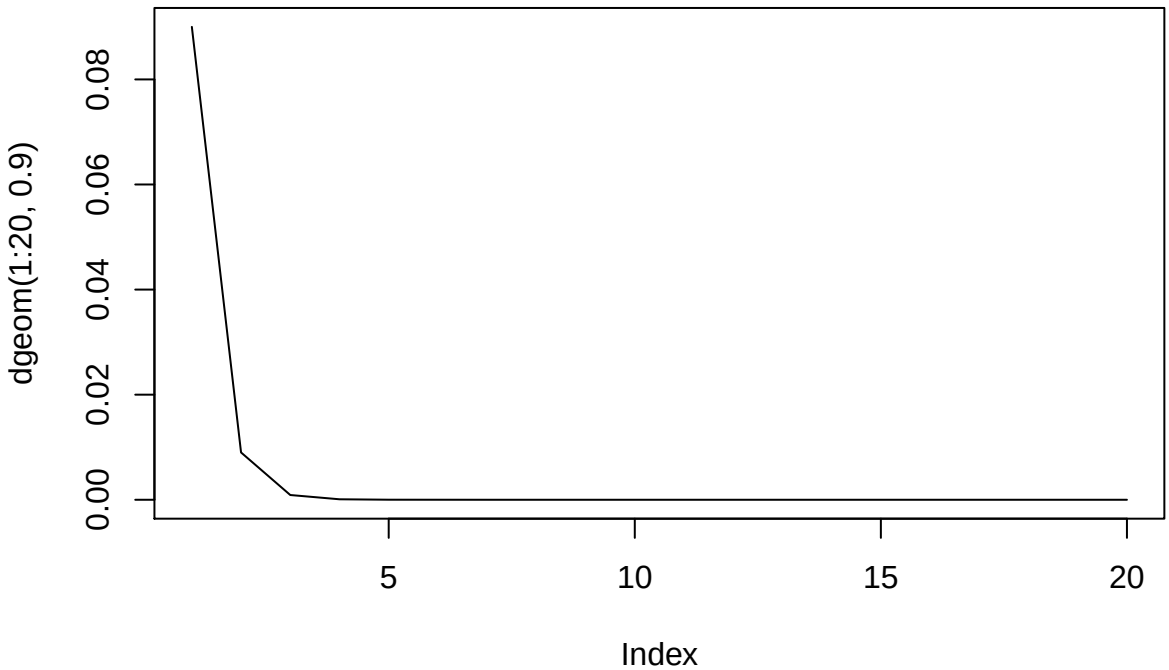
```
plot(dgeom(1:20, .5), type = 'l')
```



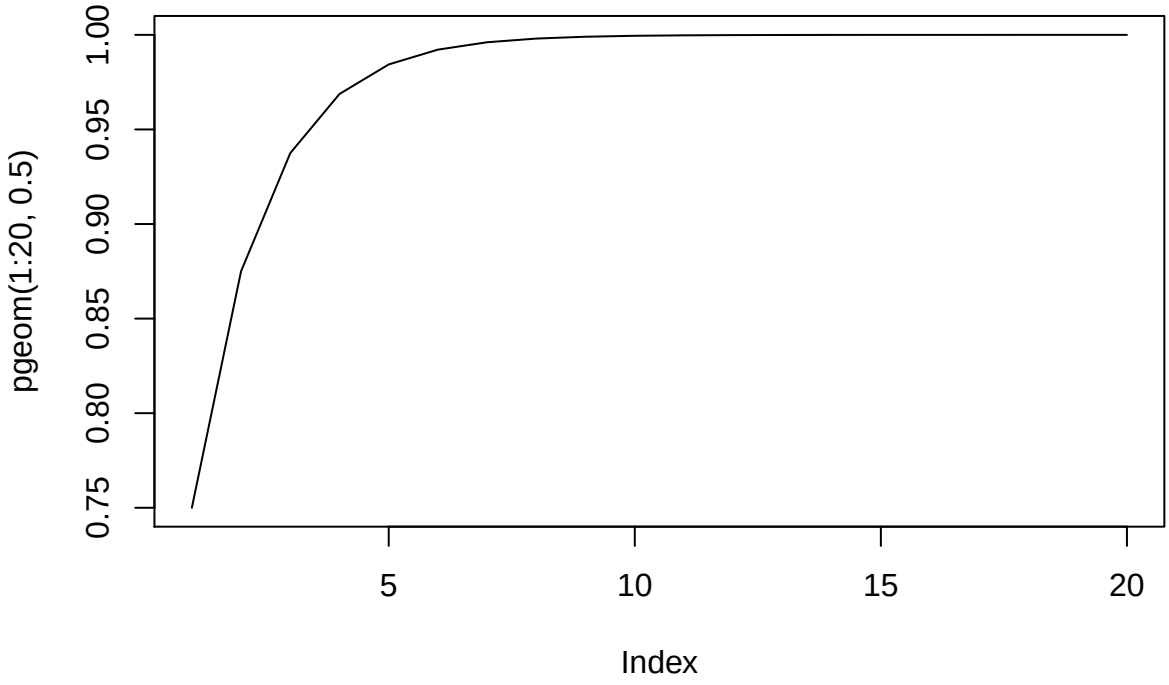
```r
plot(dgeom(1:20, .25), type = 'l')
```



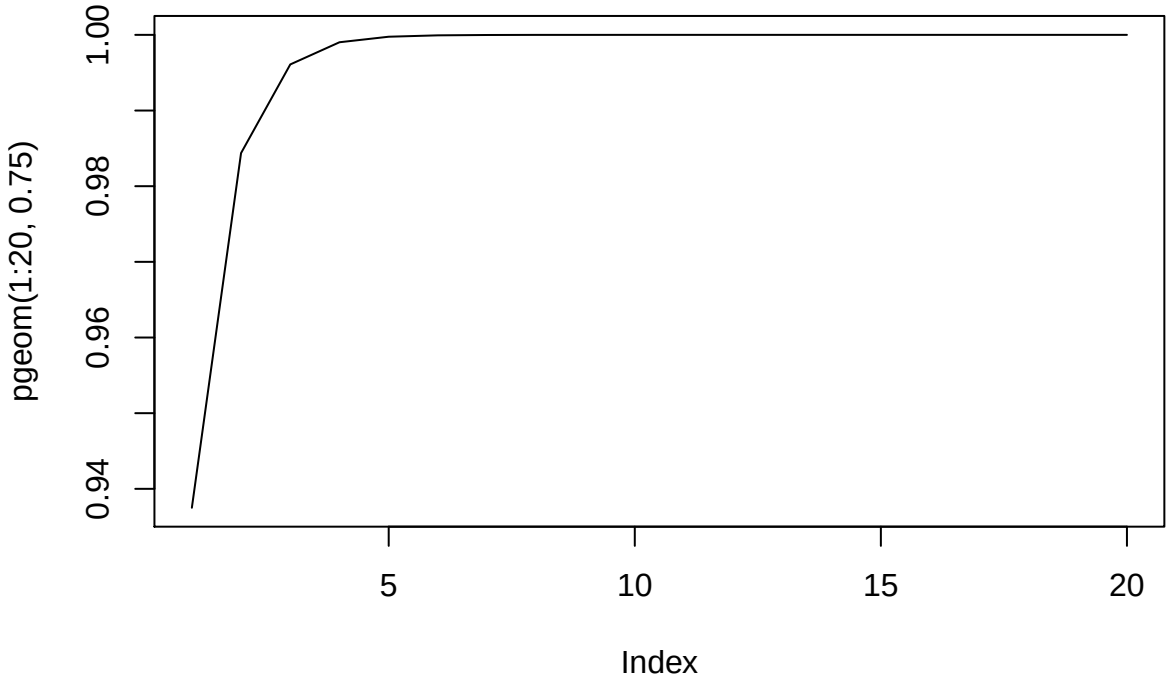
```r
plot(dgeom(1:20, .9), type = 'l')
```



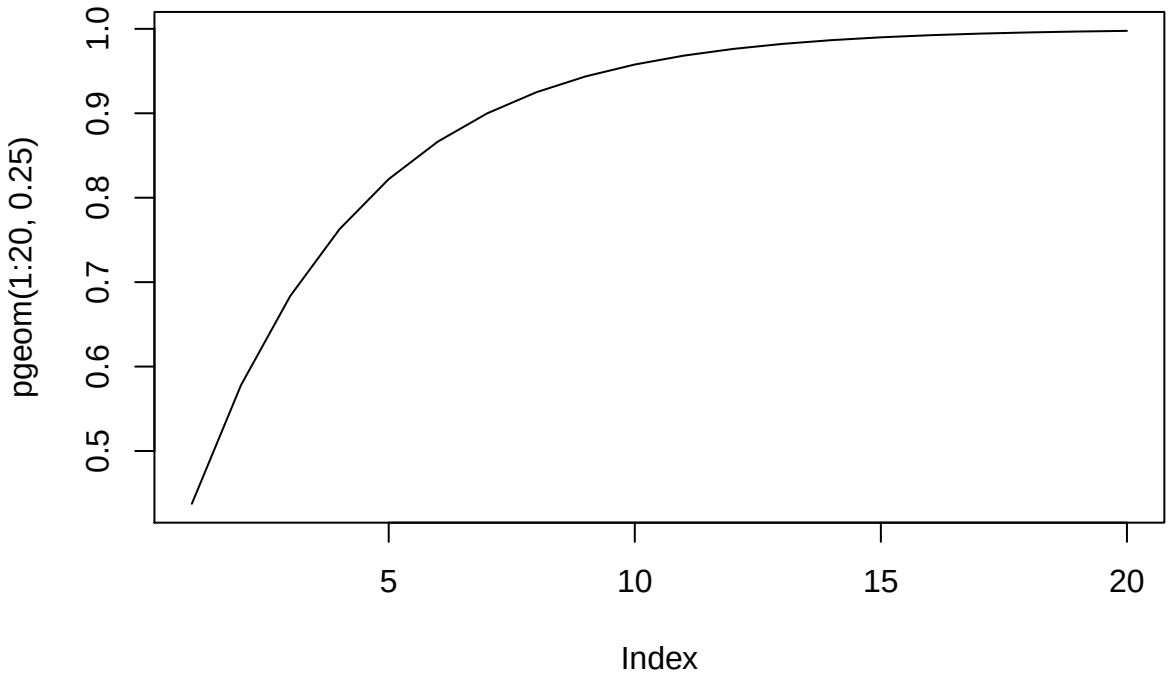
```r
plot(pgeom(1:20, .5), type= 'l')
```



```r
plot(pgeom(1:20, .75), type= 'l')
```



```r
plot(pgeom(1:20, .25), type= 'l')
```



Use previous plot as reference

```r
pgeom(5, .25)
```

```
## [1] 0.8220215
```

```r
qgeom(.5, .25)
```

```
## [1] 2
```

```r
qgeom(.8, .25)
```

```
## [1] 5
```

```r
rgeom(10, .5)
```

```
##  [1] 0 1 3 1 6 0 0 0 0 0
```

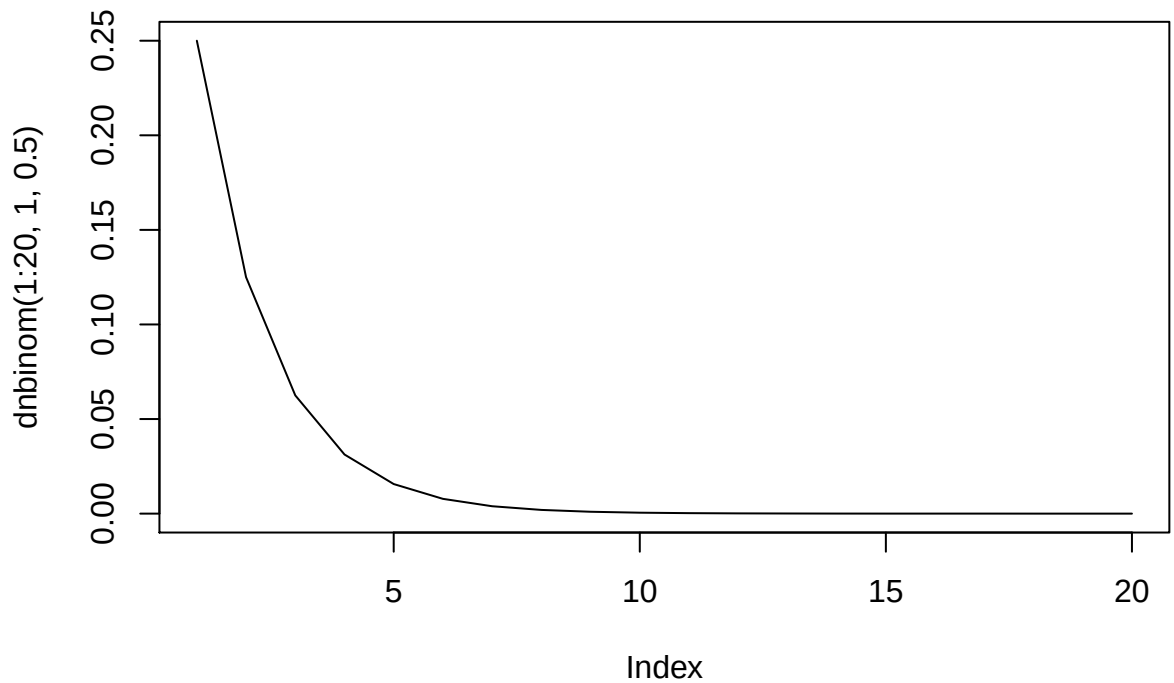
```r
rgeom(10, .25)
```

```
##  [1] 0 3 2 1 3 4 6 0 4 1
```

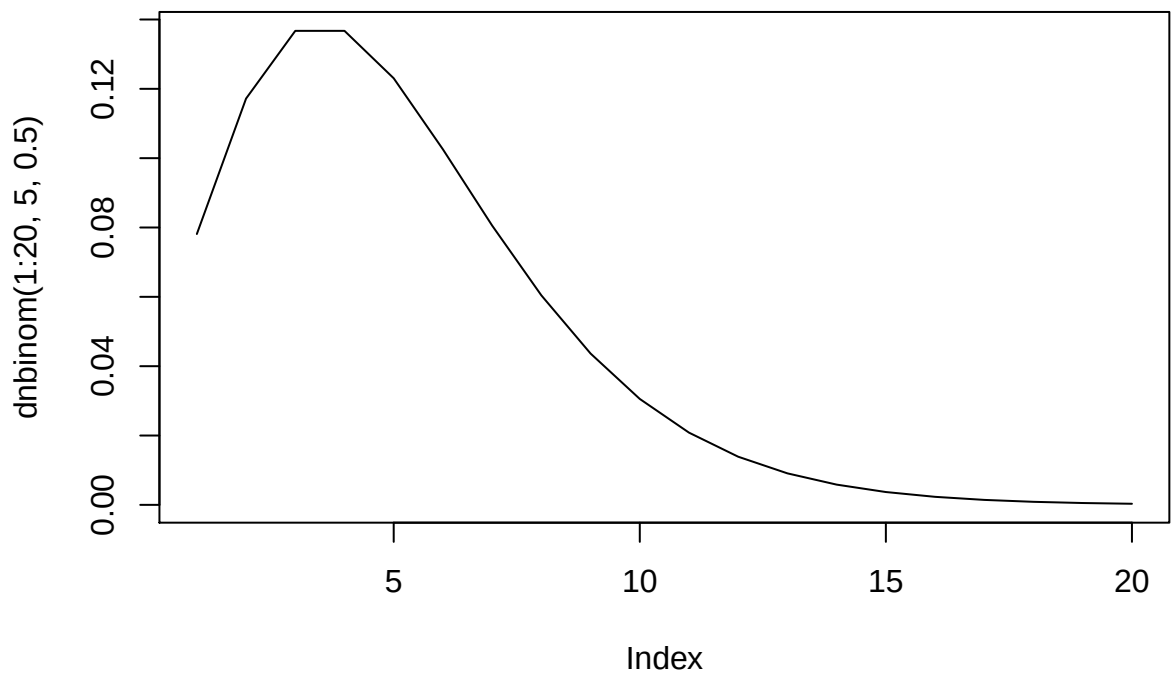
NegBinomial

Note that the geometric distribution is just the negative binomial distribution with size = 1

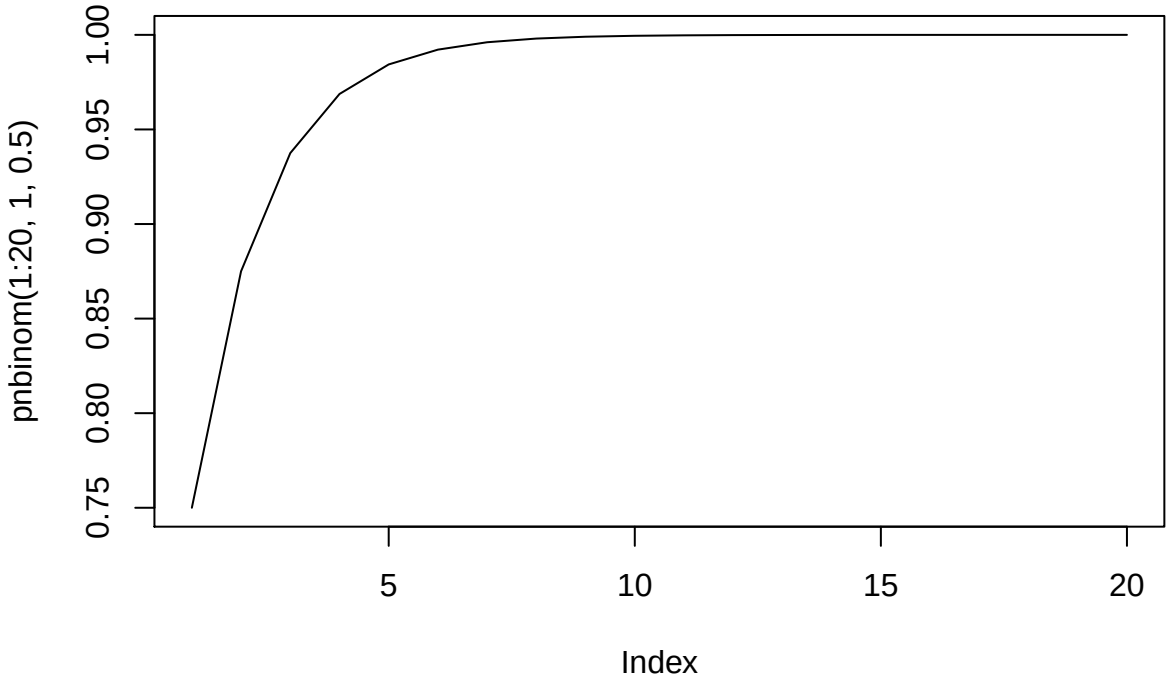
```
plot(dnbinom(1:20,1,.5), type = 'l')
```



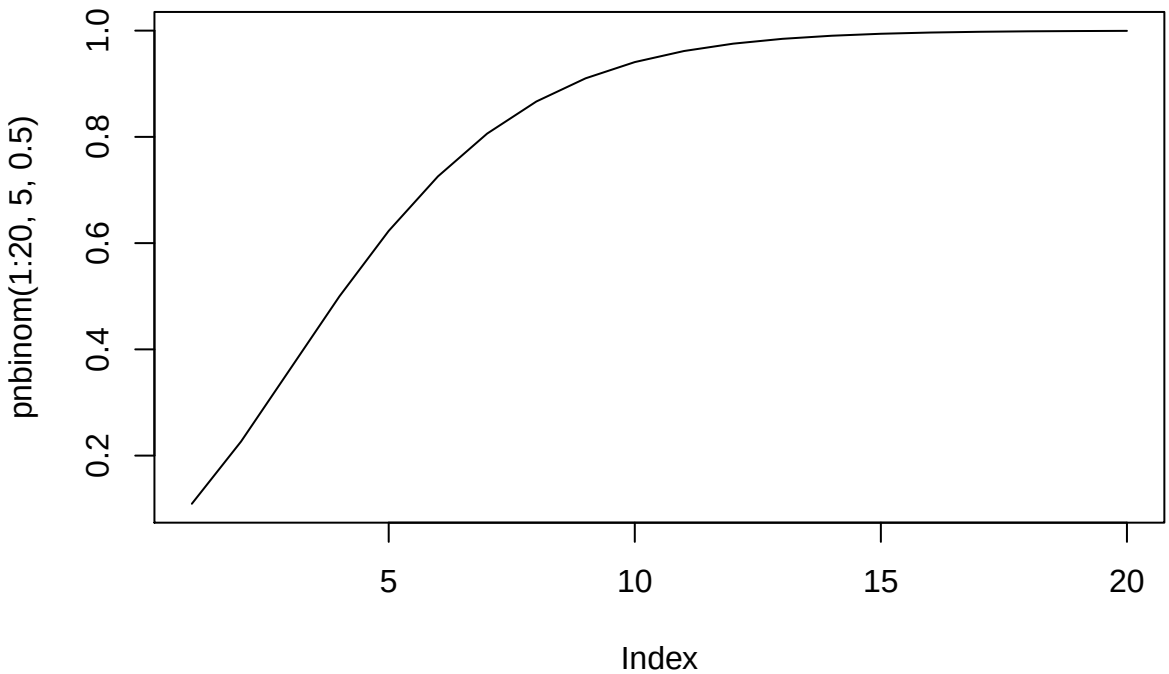
```r
plot(dnbinom(1:20, 5, .5), type = 'l')
```



```r
plot(pnbinom(1:20,1,.5),type = 'l')
```



```r
plot(pnbinom(1:20,5,.5),type = 'l')
```




```r
qnbinom(.5, 5, .5)
```

```
## [1] 4
```

```r
rnbinom(10, 5, .5)
```

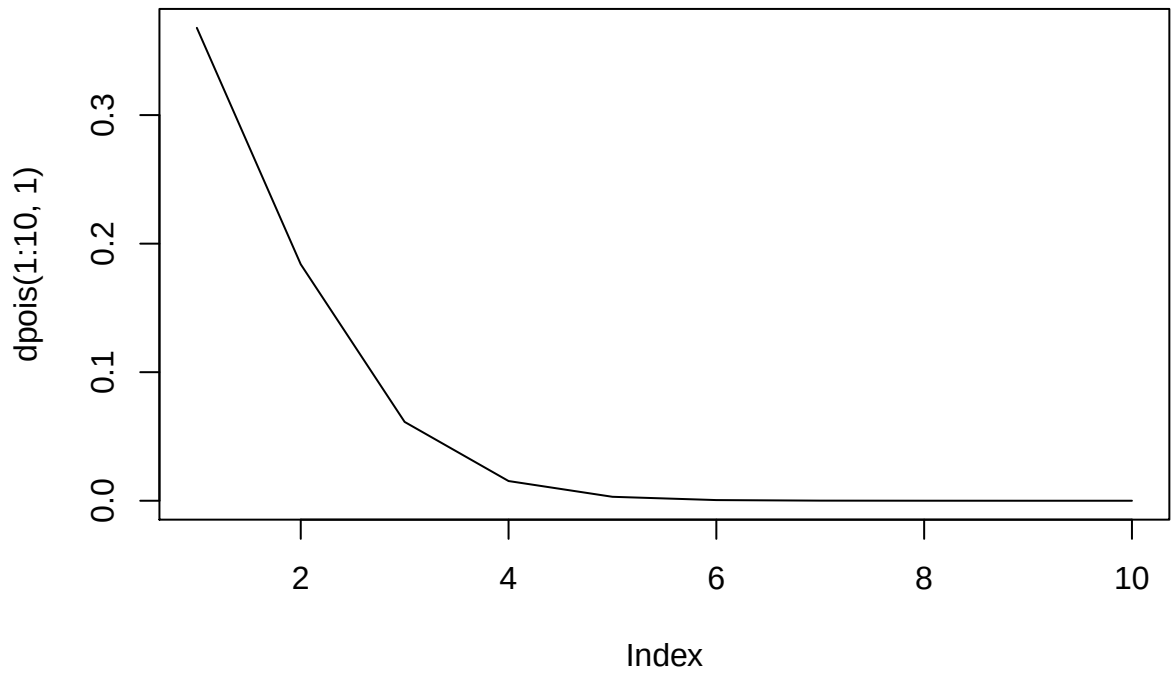
```
##  [1]  2  6 13  4  3  1  0  3  1  3
```

```r
rnbinom(10, 1, .5)
```

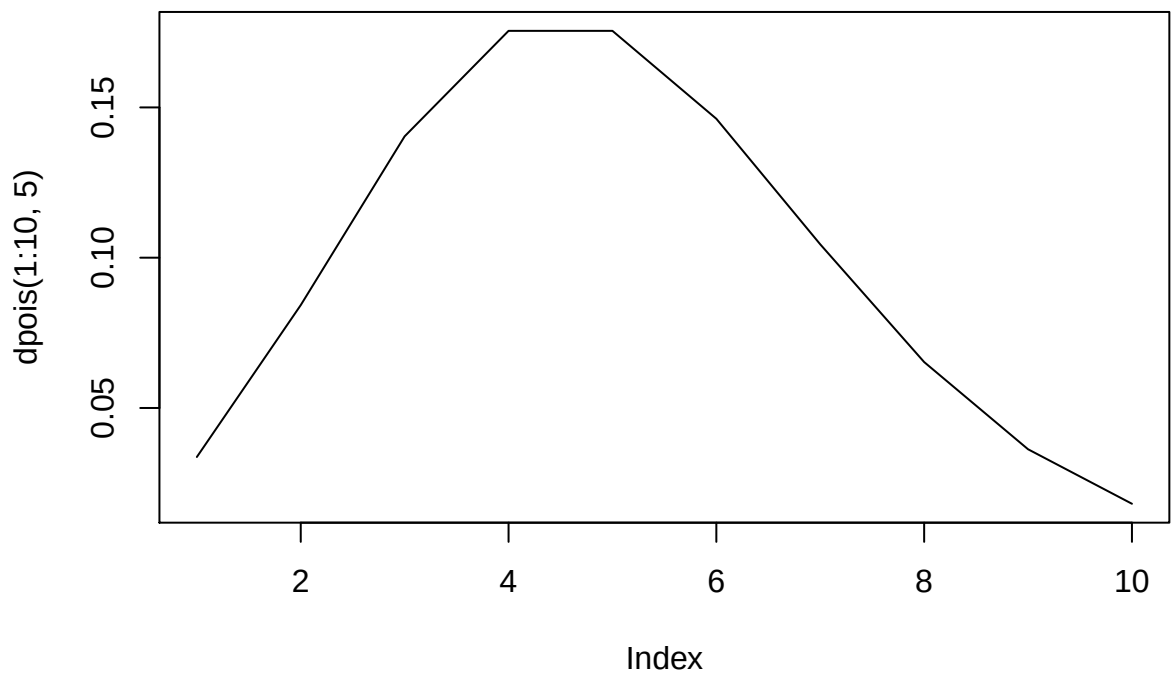
```
##  [1] 2 0 1 1 1 5 3 2 0 2
```

Poisson

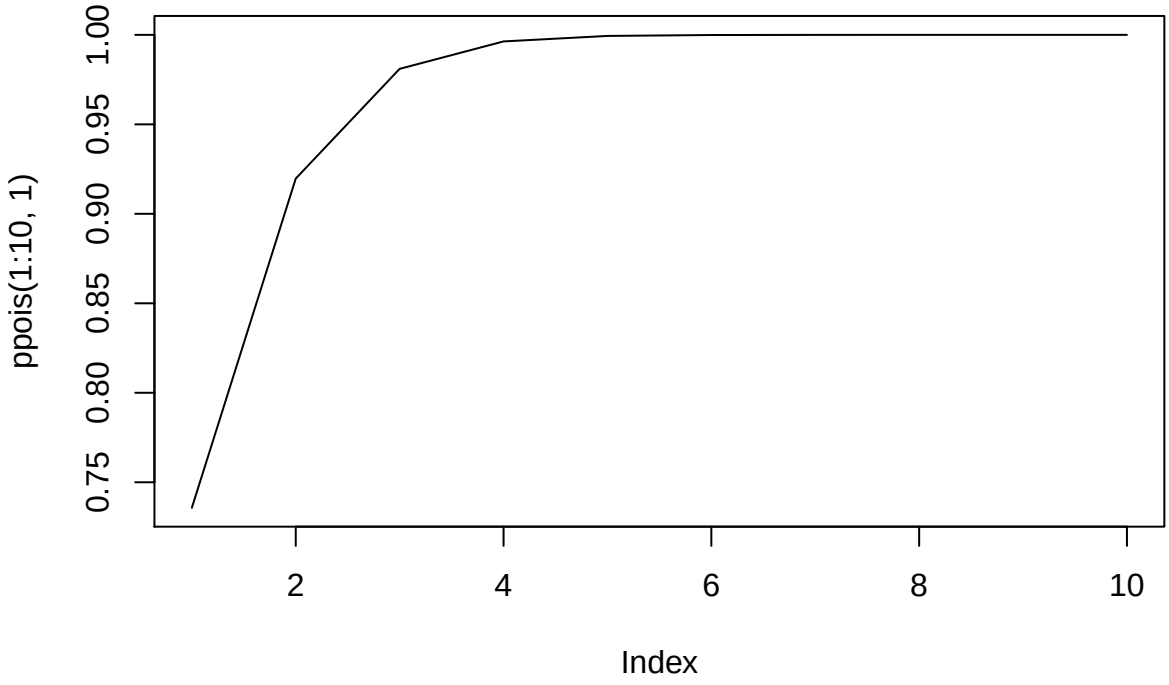
```
plot(dpois(1:10, 1), type = 'l')
```



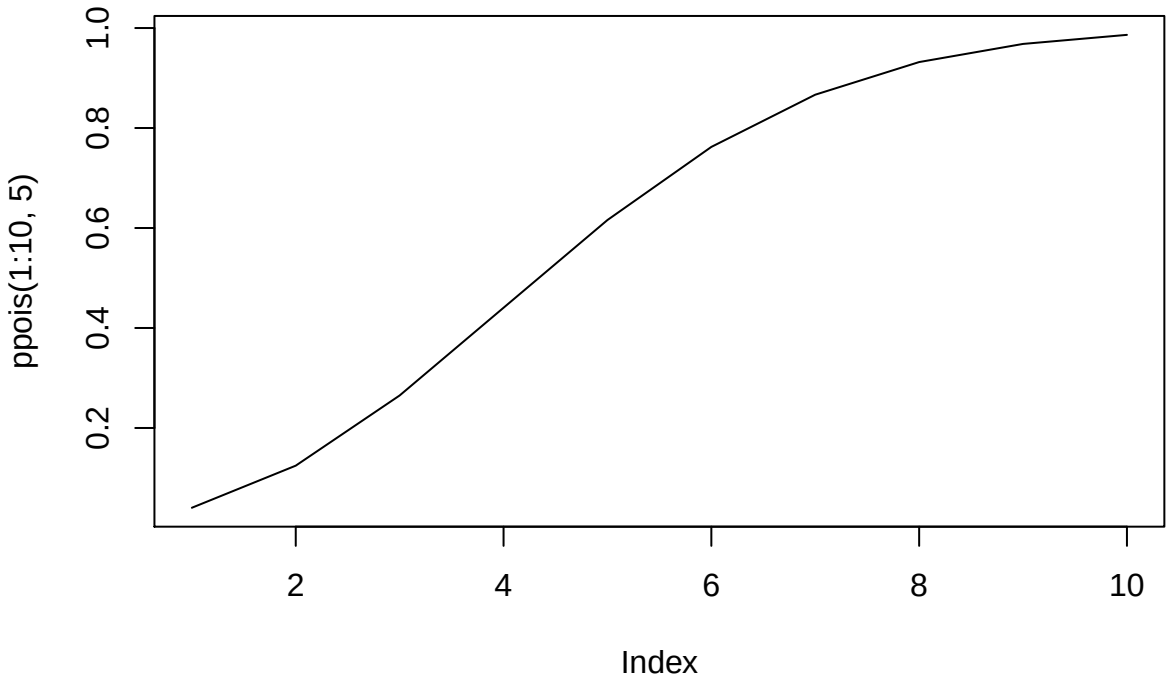
```r
plot(dpois(1:10, 5), type = 'l')
```



```r
plot(ppois(1:10, 1), type = 'l')
```



```r
plot(ppois(1:10, 5), type = 'l')
```




```r
qpois(.5, 1)
```

```
## [1] 1
```

```r
qpois(.5, 5)
```

```
## [1] 5
```

```r
qpois(.5, 1000)
```

```
## [1] 1000
```

```r
rpois(10,1)
```

```
##  [1] 1 2 2 3 0 1 0 1 0 2
```

```r
rpois(10,5)
```

```
##  [1] 4 7 4 9 4 5 4 5 4 4
```

Using Discrete Binomial to simulate Factor variable

```r
rbinom(100,1,.5)
```

```
##   [1] 0 0 0 0 1 0 1 1 0 1 1 0 1 1 0 1 0 0 0 0 1 1 1 1 0 1 0 0 1 0 0 0 0 1 0
```

```
##  [36] 0 0 1 1 0 1 1 1 0 0 1 1 1 0 0 1 0 1 1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 1 0
##  [71] 0 0 1 1 1 1 1 0 0 0 0 0 1 1 0 0 1 1 1 1 1 0 1 0 0 0 1 0 0 0
```

Use the factor function to take rbinom output (vector of 0s and 1s) and convert to a vector of "M"s and "F"s representing Male and Female

```
Sex <- factor(rbinom(100,1,.5), levels = c(0,1), labels = c('F','M'))
```

```
Sex
```

```
##   [1] F M F M M M M F M F F F F M M F F F M M F M M M M F M M F M F F F M F
##  [36] M F M F F M M F M M F F M M M F F F M F M F F M F F F F M F F F M M M
##  [71] M M F M F M F M F M F F M F M F F F F F F F M F M F M F M F
## Levels: F M
```

Use rnorm to simulate a vector of ages

```
Age <- rnorm(100, mean = 40, sd = 10)
```

```
Age
```

```
##   [1] 41.69201 35.25003 37.47576 19.74938 46.52348 38.54867 47.94022
##   [8] 44.85194 46.16467 29.93309 34.05868 30.77629 39.31269 31.89475
##  [15] 37.40060 51.44826 44.65666 26.11234 30.95902 27.18533 40.10256
##  [22] 36.21875 43.21808 40.91426 31.09344 47.91053 35.84791 40.18176
##  [29] 35.00213 31.45932 37.50782 49.34084 34.89188 44.74928 58.44867
##  [36] 33.04629 50.80889 42.66895 45.02163 55.45095 32.42954 38.27291
##  [43] 27.31949 17.65200 41.30042 47.69917 44.78357 34.32962 51.99942
##  [50] 30.99391 29.37210 37.32132 41.27588 45.67688 48.03242 52.23069
##  [57] 32.17172 44.25850 40.63368 21.72944 36.49411 52.71915 43.93670
##  [64] 19.22928 48.91894 68.61394 21.54492 67.58417 65.24728 40.40148
##  [71] 28.36441 36.36529 37.59109 37.30793 45.87038 36.93771 44.24256
##  [78] 40.06964 20.15687 42.98004 35.79023 26.48691 30.33955 43.11293
##  [85] 42.35627 52.44453 41.28682 39.55877 39.83460 44.25037 40.56622
##  [92] 48.68176 42.18547 49.25515 36.42598 27.56197 35.66942 41.96214
##  [99] 44.78934 61.08039
```

```
summary(Sex)
```

```
##  F  M
## 54 46
```

```
summary(Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   17.65   34.26   40.14   39.94   44.89   68.61
```

```
df <- data.frame(Sex, Age)
df
```

```
##     Sex      Age
## 1     F 41.69201
## 2     M 35.25003
## 3     F 37.47576
## 4     M 19.74938
## 5     M 46.52348
## 6     M 38.54867
```

```
## 7     M 47.94022
## 8     F 44.85194
## 9     M 46.16467
## 10    F 29.93309
## 11    F 34.05868
## 12    F 30.77629
## 13    F 39.31269
## 14    M 31.89475
## 15    M 37.40060
## 16    F 51.44826
## 17    F 44.65666
## 18    F 26.11234
## 19    M 30.95902
## 20    M 27.18533
## 21    F 40.10256
## 22    M 36.21875
## 23    M 43.21808
## 24    M 40.91426
## 25    M 31.09344
## 26    F 47.91053
## 27    M 35.84791
## 28    M 40.18176
## 29    F 35.00213
## 30    M 31.45932
## 31    F 37.50782
## 32    F 49.34084
## 33    F 34.89188
## 34    M 44.74928
## 35    F 58.44867
## 36    M 33.04629
## 37    F 50.80889
## 38    M 42.66895
## 39    F 45.02163
## 40    F 55.45095
## 41    M 32.42954
## 42    M 38.27291
## 43    F 27.31949
## 44    M 17.65200
## 45    M 41.30042
## 46    F 47.69917
## 47    F 44.78357
## 48    M 34.32962
## 49    M 51.99942
## 50    M 30.99391
## 51    F 29.37210
## 52    F 37.32132
## 53    F 41.27588
## 54    M 45.67688
## 55    F 48.03242
## 56    M 52.23069
## 57    F 32.17172
## 58    F 44.25850
## 59    M 40.63368
## 60    F 21.72944
```

```
## 61    F 36.49411
## 62    F 52.71915
## 63    F 43.93670
## 64    M 19.22928
## 65    F 48.91894
## 66    F 68.61394
## 67    F 21.54492
## 68    M 67.58417
## 69    M 65.24728
## 70    M 40.40148
## 71    M 28.36441
## 72    M 36.36529
## 73    F 37.59109
## 74    M 37.30793
## 75    F 45.87038
## 76    M 36.93771
## 77    F 44.24256
## 78    M 40.06964
## 79    F 20.15687
## 80    M 42.98004
## 81    F 35.79023
## 82    F 26.48691
## 83    M 30.33955
## 84    F 43.11293
## 85    M 42.35627
## 86    F 52.44453
## 87    F 41.28682
## 88    F 39.55877
## 89    F 39.83460
## 90    F 44.25037
## 91    F 40.56622
## 92    F 48.68176
## 93    M 42.18547
## 94    F 49.25515
## 95    M 36.42598
## 96    F 27.56197
## 97    M 35.66942
## 98    F 41.96214
## 99    M 44.78934
## 100   F 61.08039
```